

Drugs acting on ANS

Objective

After completing this chapter the students are expected to:

- ✓ Identify the different classes of drugs affecting the autonomic nervous system (autonomic drugs)
- ✓ Differentiate drugs acting on adrenergic system and cholinergic system
- ✓ Discuss therapeutic uses, side effects and contraindications of commonly used autonomic drugs.

Over view of pharmacology of Autonomic drugs

Autonomic NS

- The term autonomic refers to “automatic or self – regulating”
- The **autonomic nervous system (ANS)** is the branch of the **peripheral nervous system** that deals with activities that are not under direct conscious control; i.e., the so-called involuntary organs (visceral organs, exocrine glands, blood vessels, heart). such as cardiac output, blood flow to various organs, and digestion
- The ANS consists of two main divisions:
 - ✓ The **sympathetic nervous system** arises from the **thoracic and lumbar spinal cord**, and is sometimes designated as the thoracolumbar division.
 - ✓ The **parasympathetic nervous system** arises from the **cranial** nerve nuclei of the brainstem and from nerves coming from the sacral spinal cord, and is sometimes designated as the craniosacral division.

Autonomic NS...

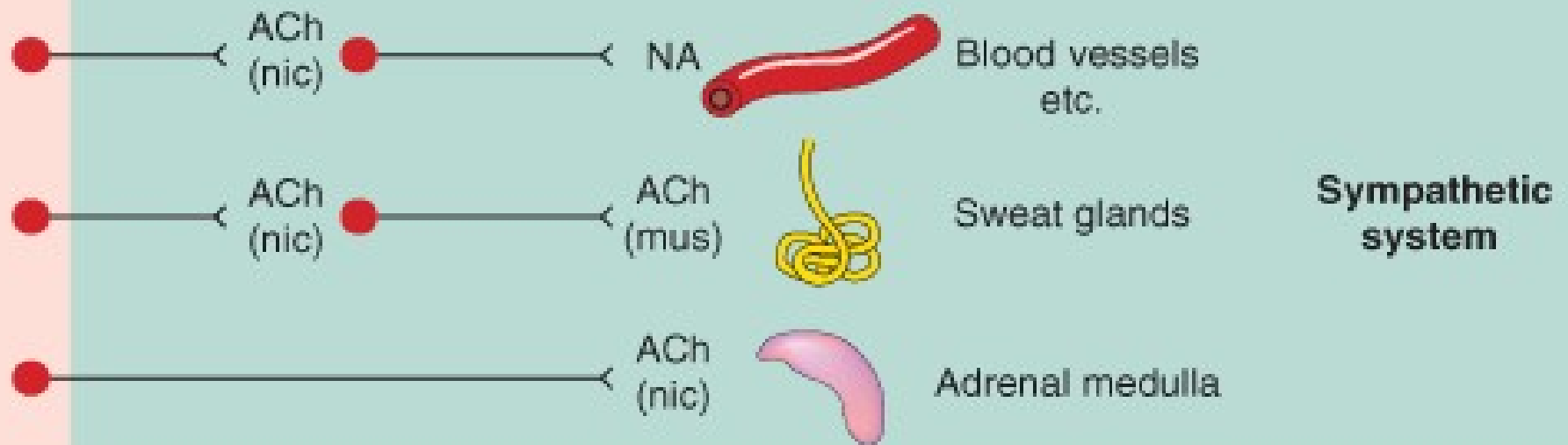
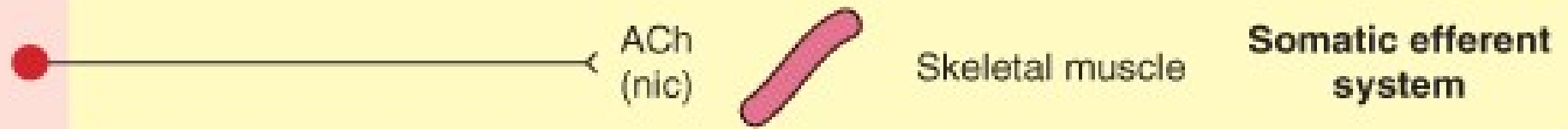
- In many cases, the autonomic effector organs are innervated by both the sympathetic and parasympathetic nervous systems. Generally, the actions of the two systems are opposite, and the regulation of the two systems is carried out in a reciprocal fashion. For example, sympathetic stimulation of the heart results in an increase in heart rate (and contraction), while parasympathetic stimulation reduces heart rate.
- The nerve pathway in the ANS to a given tissue consists of two neurons. The preganglionic nerve comes from the spinal cord and synapses onto the postganglionic fiber, which then innervates the target organ. The point of contact of the two neurons is called the *ganglion*.

Autonomic NS...

- Peripheral nervous system provides a double set of nerve fibers:
 - Sympathetic (adrenergic)
 - Fight or flight action
 - Exit from thoracic and lumbar regions
 - Neurotransmitter- norepinephrine
 - Preganglionic neuron is shorter than postganglionic neuron
 - Adrenergic receptors- α_1 , α_2 , β_1 , β_2 , β_3
 - Parasympathetic (cholinergic)
 - Rest or digest
 - Exit from cranial and sacral portions
 - Neurotransmitter- acetylcholine
 - Preganglionic neuron is longer than postganglionic neuron
 - Cholinergic receptors- nicotinic and muscarinic receptors.

Neurotransmitters

- There are two important neurotransmitters in the autonomic nervous system. These are acetylcholine and noradrenaline (norepinephrine).
- Acetylcholine is a neurotransmitter which is released after stimulation of the parasympathetic nervous system to act on effector organs (cells) to elicit their response, but it also acts as a neurotransmitter:
 - At the ganglia of both sympathetic and parasympathetic NS,
 - At postganglionic sympathetic nerve endings to blood vessels of skeletal muscles and sweat glands (eccrine),
 - At the neuromuscular junction of skeletal muscles (somatic motor fibers to skeletal muscle),
 - Between some neurons in the CNS, and
 - At preganglionic nerve endings to the adrenal medulla.



Steps in Neurotransmission

- There are 5 key steps in neurotransmission
 - **Synthesis**
 - **Storage**
 - **Release**
 - **Action on receptors.**
 - **Termination of the action of the transmitter**

Functional differences

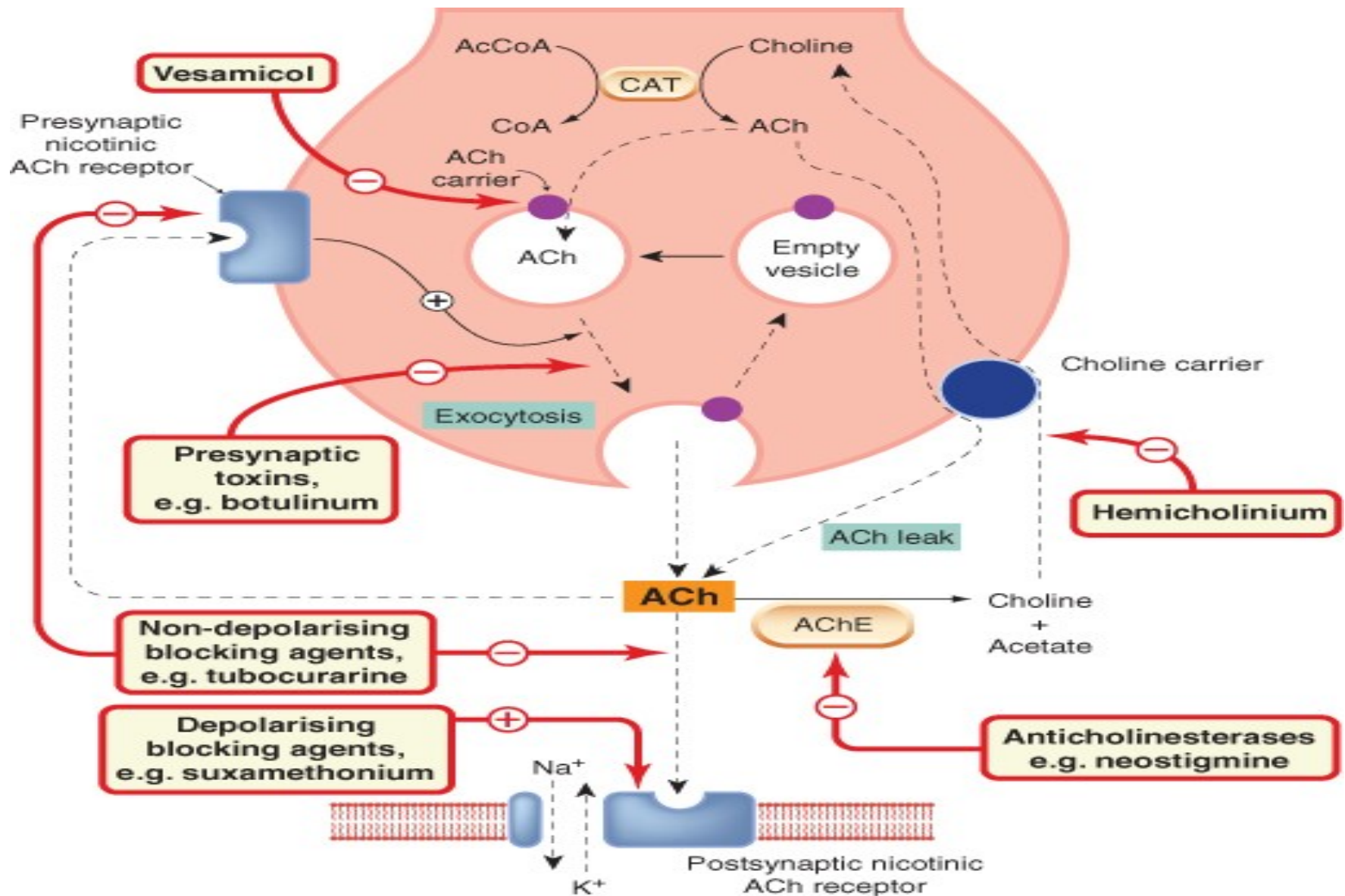
Effector Organ	Sympathetic	Parasympathetic
Heart	Heart Rate, force of Contraction, Conduction Velo. increased	decreased
Lungs	Relaxation of bronchial muscle	Contraction
Arterioles of skin, mucosa & abdominal viscera	constriction	-
Arterioles of skeletal muscles, heart	dilation	constriction
Gastrointestinal tract	GI Motility and secretions decreased	increased
Sweat glands secretion	increased	decreased
Radial muscle of iris	Contraction- mydriasis	-
Sphincter muscle of iris	-	Contraction- miosis
Ciliary muscle	Relaxation-far vision	Contraction - near vision
Secretions	decrease(GI, salivary, Lacrimal, respiratory, nasopharyngeal)	increase

Classifications of drugs acting on ANS

- Cholinergic (parasympathomimetic)
- Cholinergic blockers (parasympatholytic)
- Adrenergic (sympathomimetic)
- Adrenergic blockers (sympatholytic)

**Parasympathomimetic
(Cholinergic Agents)**

Cholinergic Transmission



Cholinergic Agents

- Cholinergic drugs are also called parasympathomimetics because their effect mimics the effect of parasympathetic nerve stimulation.
- Administration of these drugs will result in an increase in the parasympathetic activities in the systems innervated by cholinergic nerves.

There are two groups of cholinergic drugs:

1. *Direct-acting: bind to and activate muscarinic or nicotinic receptors (mostly both)*
2. *Indirect-acting: inhibit the action of acetylcholinesterase enzyme*

The actions of acetylcholine may be divided into two main groups: -

1. Nicotinic actions- those produced by stimulation of all autonomic ganglia and the neuromuscular junction
2. Muscarinic actions- those produced at postganglionic cholinergic nerve endings

Cholinergic Agents

Direct acting

1.Choline esters

- Acetylcholine
- Bethanechol
- Carbachol

2. Naturally Occurring Alkaloids

- Nicotine
- Muscarine
- Pilocarpine

Indirect acting

1.Short acting(reversible)

- edrophonium

2.Medium acting(reversible)

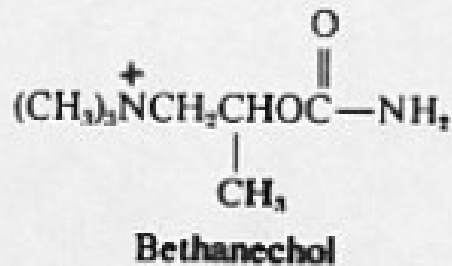
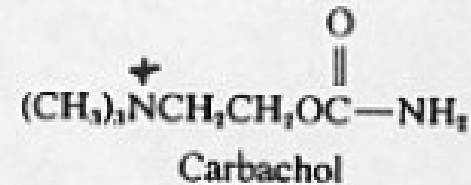
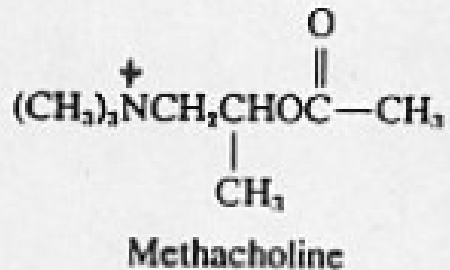
- Neostigmine
- Pyridostigmine
- physostigmine

3.long acting(irreversible)

- Organophosphates
- ecothiophate

Direct-acting cholinergic drugs(agonist)

1.Choline esters



1.Choline esters contd.

- **Acetylcholine**

- Ester of acetic acid and choline
- Choline group contains a **quaternary ammonium gp** that confers high polarity (hydrophilicity)
- Cholinesterases catalyze hydrolysis of Ach to acetate and choline
 - and this is the only means of termination of transmitter action of Ach

- **There are 2 limitation in therapeutic usefulness of Ach.**

1.Choline esters contd.

- **Derivatives of Ach**

Methacholine, carbachol and bethanechol

- Are too hydrophilic to cross membranes
- Poorly absorbed & poorly distributed into the CNS because they are hydrophilic

- **Methacholine**

- Is structurally **beta-methylated** Ach which renders the drug **more selective to muscarinic receptors and resistant to cholinesterase activity**
- Increased potency and duration of action relative to Ach

1.Choline esters contd.

- **Carbachol**

- Differs from Ach only in the substitution of a carbamoyl group for the terminal methyl group of Ach
- **This renders carbachol completely resistant to degradation by cholinesterase**
- Its receptor selectivity is not improved for muscarinic Vs nicotinic

- **Bethanechol**

- Combines the addition of a methyl group and the substitution of the terminal carbamoyl group
- It is selective agonist of muscarinic receptors and is resistant to degradation by cholinesterases

2. Naturally Occurring Alkaloids

Nicotine

- Nicotine derived from tobacco
- Mimics the actions of acetylcholine at nicotinic sites
- Has affinity for nicotinic receptors than muscarinic receptors.

Pilocarpine

- Has affinity for muscarinic receptors

Muscarine

- Comes from certain mushrooms
- Muscarine mimics the actions of acetylcholine at smooth muscles, cardiac muscles, and glands
- ✓ **Pilocarpine** and **nicotine** are tertiary amines
- ✓ **Muscarine**, a quaternary ammonium cpd

Indirect acting cholinomimetics

- There are three classes of AChEIs
 1. Reversible- Short acting
 - Eg- edrophonium, tacrine
 2. Carbamylating agents- Intermediate acting
 - Eg- Neostigmine and Physostigmine
 3. Phosphorylating agents- long acting
 - Eg. organophosphates

Therapeutic uses of reversible inhibitors of AChEIs

- **Edrophonium**- reversible and rapid renal clearance
 - Diagnosis of myasthenia gravis
- **Tacrine**-more lipophilic and has longer duration
 - For treatment of Alzheimer's disease

Carbamyl Inhibitors

- Physostigmine
- Neostigmine (N^+)
- Pyridostigmine (N^+)
 - Form moderately stable complex with AChE that lasts from 3-4 hrs
 - The carbamyl esterase linkage is hydrolysed by the esterase, but at much more slowly compared to acetylcholine
- Neostigmine and pyridostigmine are quaternary ammonium salts hence can not cross BBB
 - Not therapeutically used for treatment of atropine poisoning only physostigmine is used

Carbamylating inhibitors

Therapeutic uses

- Glaucoma
- Atony of the bladder
- Atony of the gastrointestinal tract
- Intoxication by antimuscarinic agents (use physostigmine)
- Recovery of neuromuscular function after competitive blockade of N_M receptor of skeletal muscle fibers
- Myasthenia gravis

Therapeutic Uses of phosphorylating agents

- **Malathion**
 - Insecticide, also parathion
- **Echothiophate**
 - local application to the eye for treatment of glaucoma

Organophosphate poisoning

- Source of intoxications ; pesticide use in agriculture and insecticides in the home.
- **SLUDE BAM**
/Salivation, Lacrimation, Urination, Emesis, Bradycardia/bronchoconstriction, Abdominal cramp, and Miosis.
 - Nicotinic Manifestations
 - Weakness or paralysis of skeletal muscle
 - “Depolarizing blockade” of NMJ
 - Paralysis of muscles of respiration
 - Respiratory depression and cardiovascular collapse

Therapy of organophosphate poisoning

- (1) maintenance of **vital signs—respiration** in particular may be impaired
- (2) **decontamination** to prevent further absorption -removal of all clothing and washing of the skin in cases of exposure to dusts and sprays
- (3) Atropine parenterally in large dose + **pralidoxime**(cholinestrase activator)
 - **pralidoxime** has to be given very early before the enzyme under goes aging.
- (4) benzodiazepines for seizures.

Therapeutic uses of cholinomimetic agents

1. Treatment of Glaucoma

- a progressive form of optic nerve damage associated with an increased
/> 21 mmHg/ IOP

2. Treatment of myasthenia gravis

- Muscle weakness and rapid fatigue of muscles during use are characteristics of the disease
- Cholinomimetic agents help to alleviate the weakness
 - elevating and prolonging the concentration of ACh in the synaptic cleft
 - producing a greater activation of the remaining nicotinic receptors.
- Pyridostigmine and neostigmine are the major.

3. Treatment of Urinary retention

- The drug of choice is **bethanecol**.

4. Treatment of GI disorder

- Gastric atony or paresis (delay in gastric emptying that occur mainly in diabetic patients).
- Bethanecol is drug of choice.

5. Treatment of xerostomia

- A disorder characterized by decrease in salivary and lacrimal gland secretions.
- Pilocarpine is helpful.

6. Treatment of Alzheimer's disease

- neurodegenerative disease that produces a progress loss of memory
and cognitive function.

- **Donepezil, Rivastigmine, Tacrine and Galanthamine.**

- Cross BBB to produce a reversible inhibition of AChE in the CNS.

Adverse effects of cholinomimetic agents

- Salivation
- Bronchial constriction.
- Urinary urgency
- GI hyperactivity
- Loss of accommodation
- Abdominal cramp
- Bradycardia

Contraindications

- ✓ Asthma patients
- ✓ Patients with hypotension and heart failure
- ✓ Peptic ulcer
- ✓ GI and urinary obstructive diseases.

Cholinergic blockers (Parasympatholytic)

Anticholinergics(Antimuscarinics)

- **Tertiary amines –**
atropine , homatropine,

hyoscine/scopolamine/,

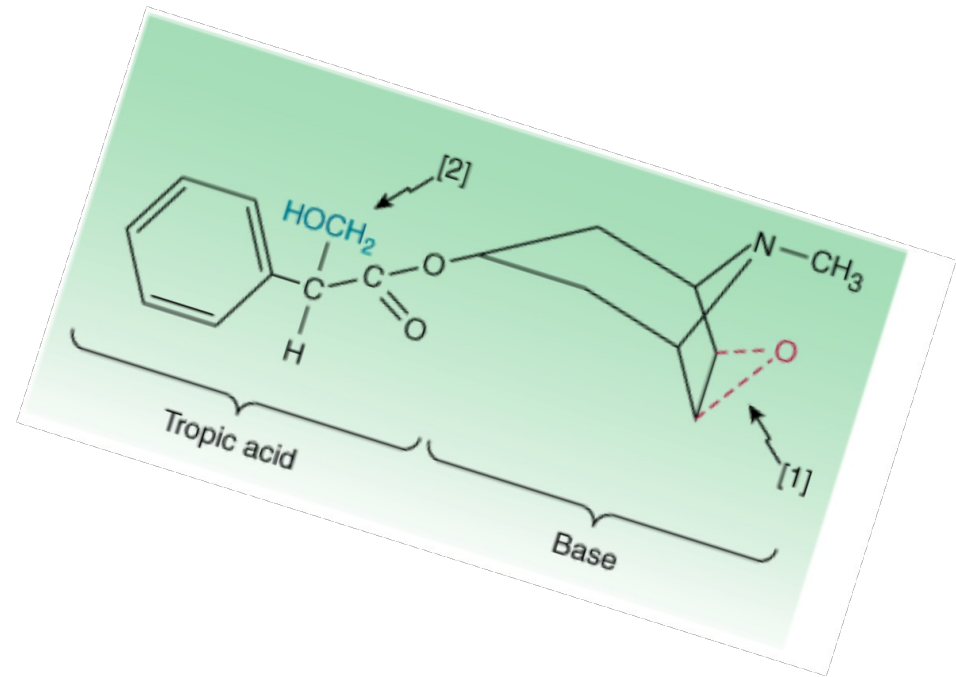
pirenzepine,
benzhexol/trihexyphendyl/,

tropicamide,cyclopentolate,

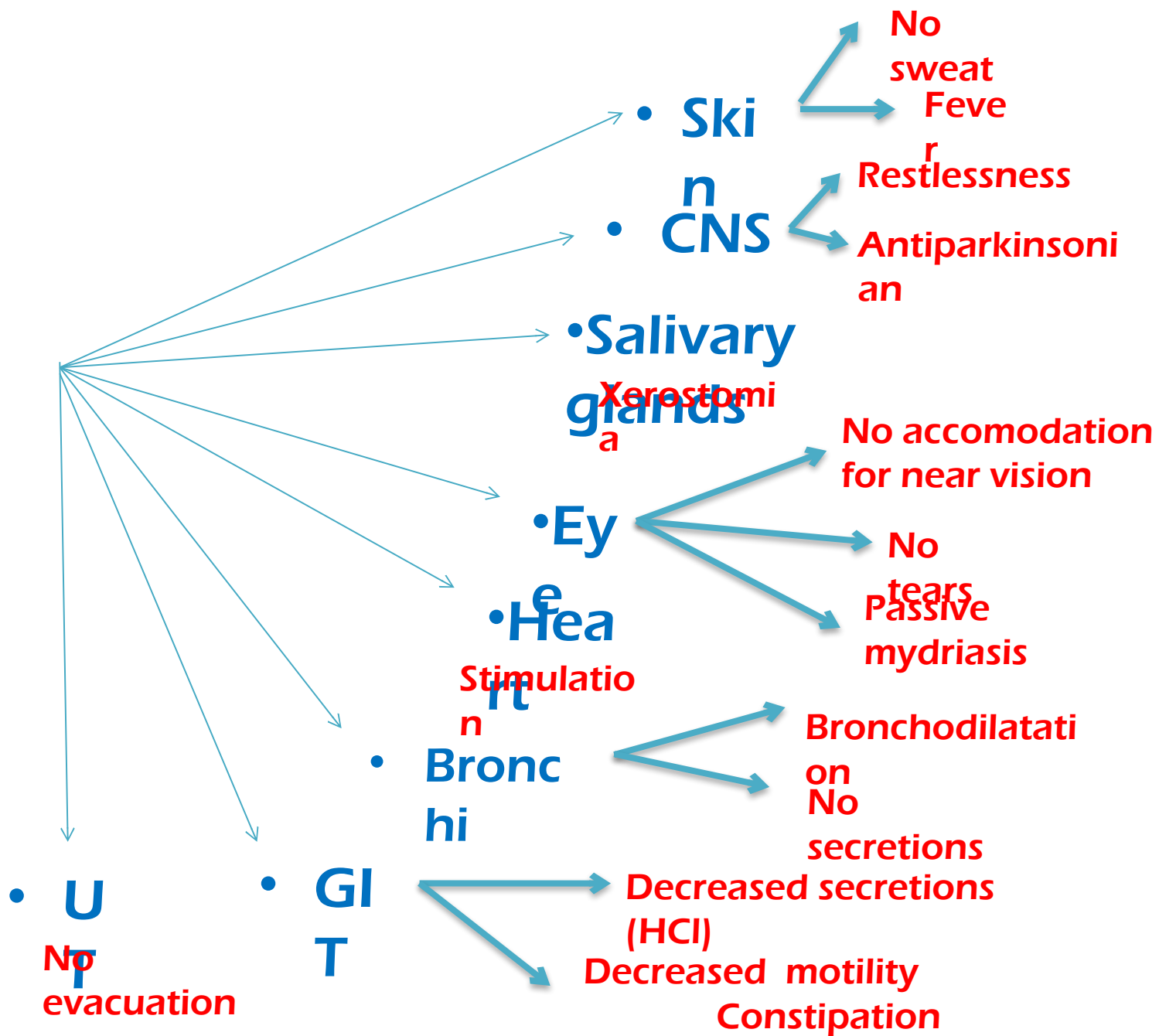
e,

- **Quaternary amines :-**

-Ipratropium bromide ,
Propantheline bromide .



Effects of Atropine



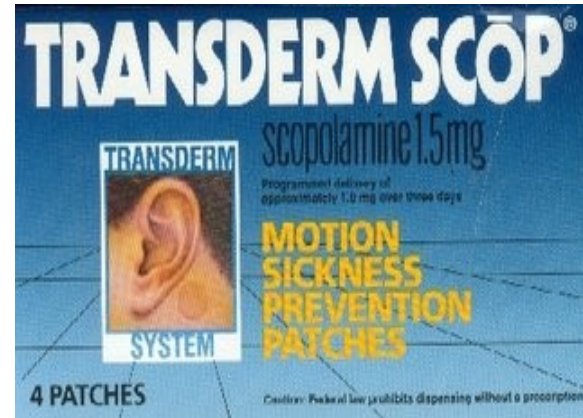
Therapeutic uses of Anticholinergics(Antimuscarinics)

✓ **Parkinson's disease**

- excess of cholinergic activity in the striatum
- Main therapy is directed toward replacement of the dopaminergic deficiency rather than blocking the cholinergic excess
- Antimuscarinics are sometimes employed for mild cases and in combination with other agents (e.g., levodopa) for treatment of advanced cases.
- **Benztropine**

Therapeutic uses...

- ✓ **Motion sickness**
 - *Scopolamine* is useful for prevention of motion sickness.
- ✓ **Uses in Ophthalmology**
 - Ophthalmoscopic examination of the retina due to mydriatic effect
- ✓ **Uses in Respiratory Disorders**
 - Atropine is used as *preoperative medication* when inhalation anesthetics such as ether are used. because atropine
 - markedly decrease airway secretions.
 - **Ipratropium**, a synthetic analog of atropine, is used as an inhalational drug in asthma.



Ipratropium

- Uses – Bronchial asthma
- Distinctiveness from atropine
 - Inhalation form
 - Quaternary ammonium salt
 - Does not cross CNS
 - Has less side effect

Adverse effects

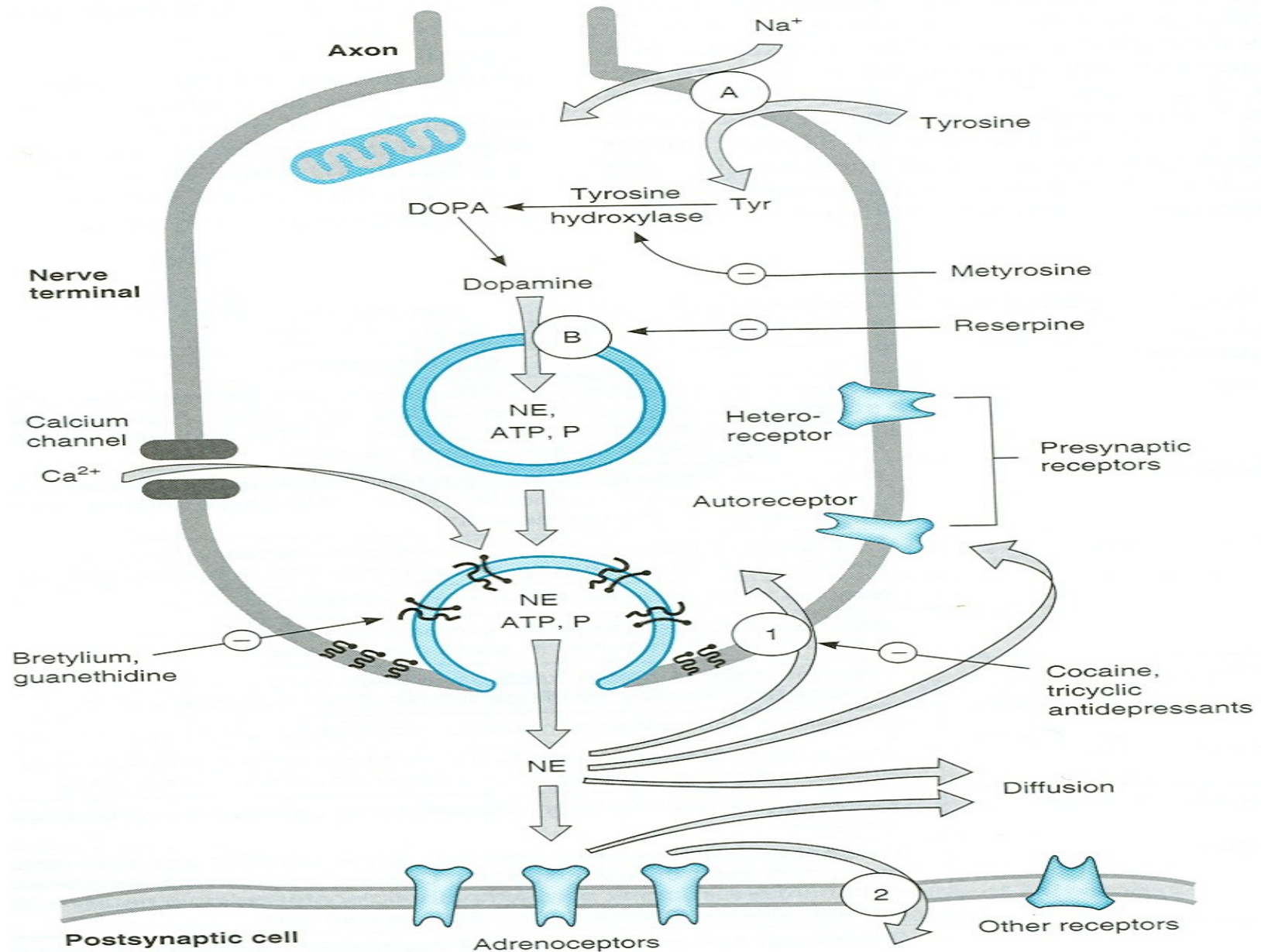
- Blurred vision
- Photophobia
- Dry mouth
- Urinary retention
- Constipation
- Elevation of IOP

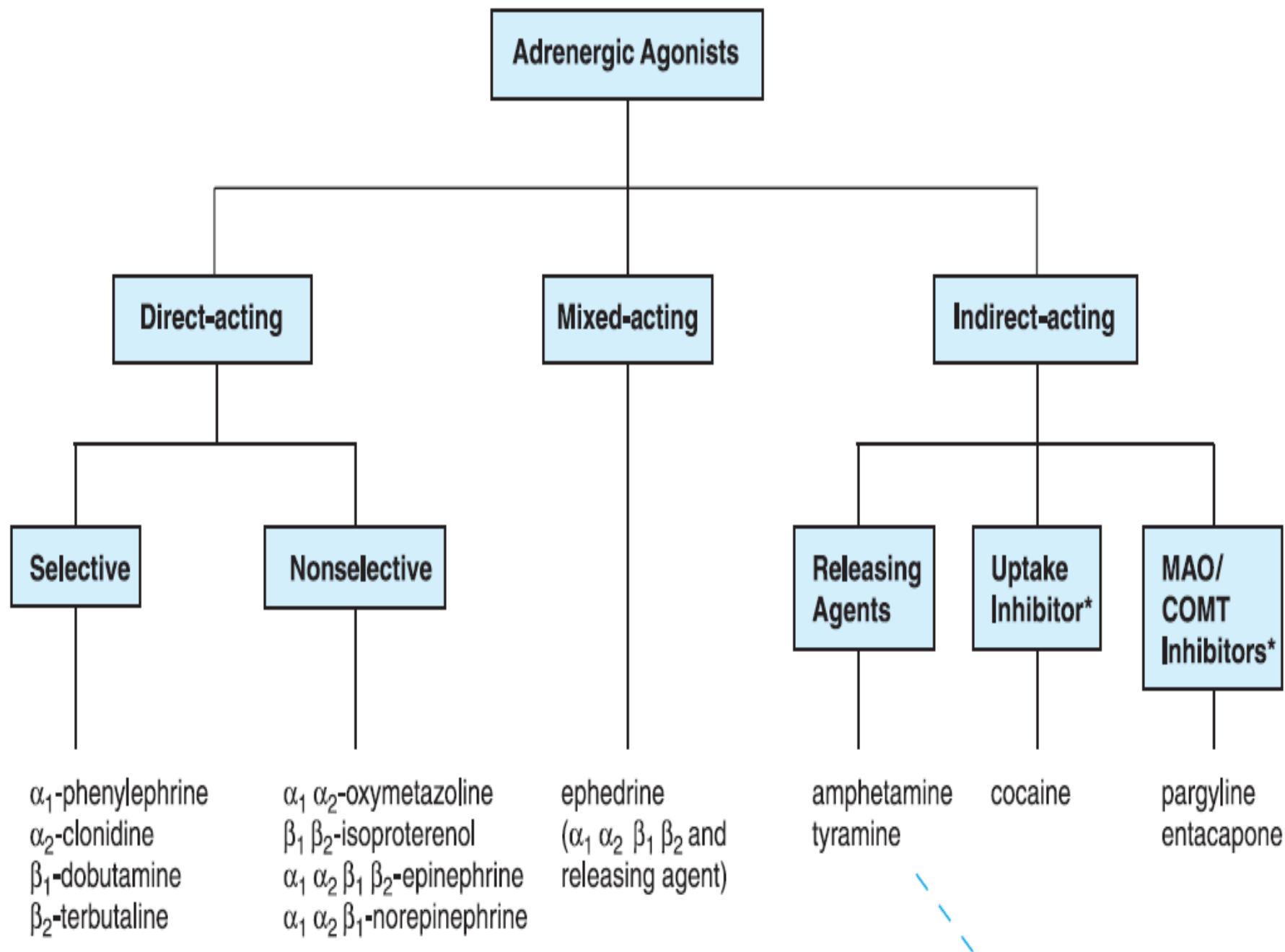
Contraindications

- Glaucoma
- Hypertrophy of the prostate gland(BPH)
- Atony of the bladder
- Atony of the GI Tract
- Myasthenia gravis
- Tachydysrhythmias
- PUD ?

Sympathomimetics (Adrenergic drugs)

Adrenergic Transmission





Adrenergic drugs

Catecholamines:

- **Natural:** Adrenaline, Noradrenaline, Dopamine
- **Synthetic:** Isoprenaline, Dobutamine

Non-Catecholamines:

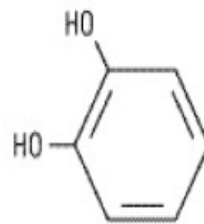
- Ephedrine, Amphetamines, Phenylephrine

Also called *sympathomimetic amines* as most of them contain an *intact or partially* substituted amino (NH₂) group

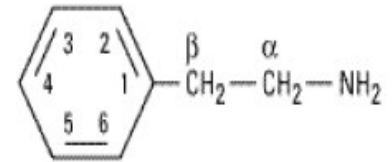
Characteristics of catecholamine's

- Have short duration of action-rapid inactivation
- Rapid onset of action
- Need parenteral administration
- Ineffective orally
- Have high potency
 - α : NE > E > Isoproterenol
 - β : Isoproterenol > E > NE
- Poor penetration in CNS

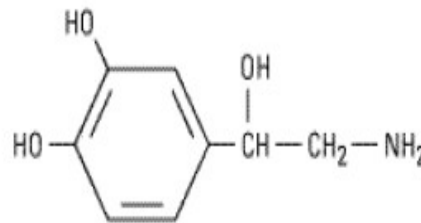
- **Catecholamines:** Cmpds containing a *catechol nucleus* (Benzene ring with 2 adjacent OH groups) and an *amine* containing side chain
- **Non-catecholamines** lack hydroxyl group



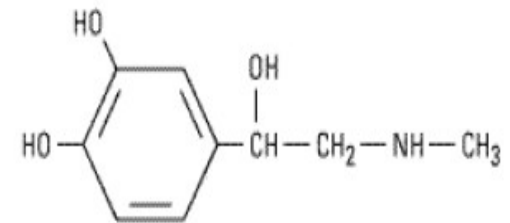
Catechol



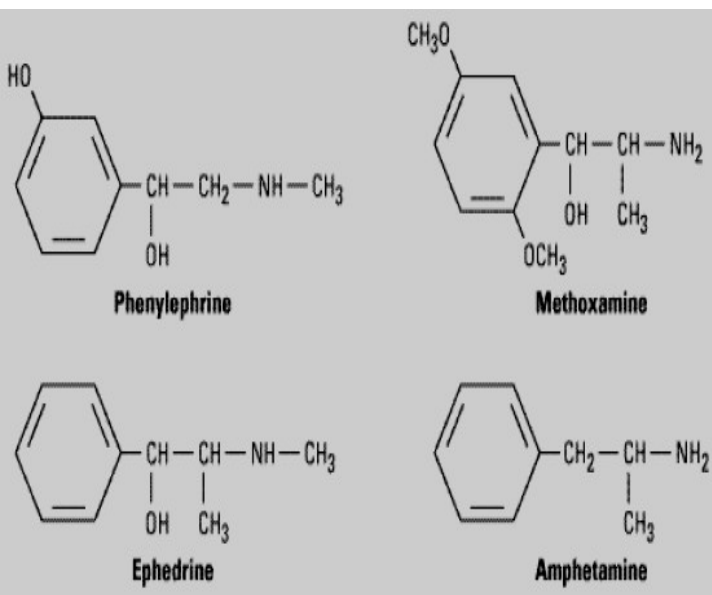
Phenylethylamine



Norepinephrine

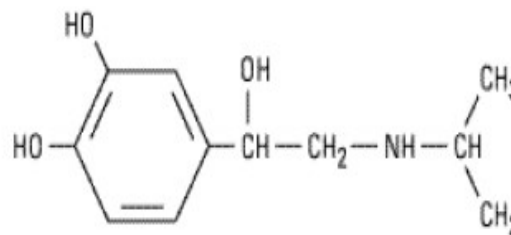


Epinephrine

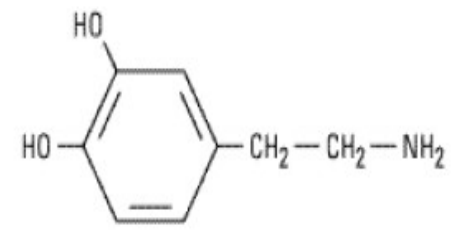


Phenylephrine

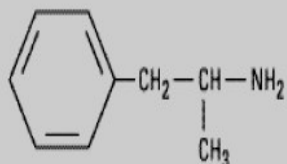
Methoxamine



Isoproterenol



Dopamine



Amphetamine

Ephedrine

Adrenergic (sympathomimetic)

<u>Drugs</u>	<u>Receptor activated</u>
Isoproterenol	B1 and B2
Adrenaline	α 1, α 2, B1 and B2
Noradrenaline	α 1, α 2 and B1
Dopamine	B1
Dobutamine	B1
Ephedrine	α 1, α 2, B1 and B2
Phenylephrine	α 1
Terbutaline, Sulbutamol	B2

Table 1: Distribution of adrenoceptor subtypes and their actions

Type	Tissue	Actions
Alpha1	Most vascular smooth muscles	Contraction
	Pupillary dilator muscle	Mydriasis
	Heart	Increase force of contraction
Alpha2	Adrenergic nerve terminals	Inhibition of transmitter release
	Platelets	Aggregation
Beta1	Heart	Increased rate and force of contraction
Beta2	Respiratory, uterine, and vascular smooth muscle	Relaxation
	Human liver	Glycogenolysis
Beta3	Fat cells	Lipolysis

Site of α_1 – adrenoceptors & effects of their stimulation

- In vascular smooth muscle.
 - α_1 stimulation cause VC :
 - **Vasoconstriction** in the skin & viscera cause increase PVR causing increase BP
 - **Vasoconstriction** in the nasal blood vessels cause **relief congestion**
- In the radial muscle of iris.
 - α_1 stimulation causes contraction of the radial muscle causing mydriasis (**dilation of the pupil**)
- In the smooth muscle of the sphincters of GIT.
 - α_1 stimulation cause contraction of all sphincters.
 - In the smooth muscle of internal sphincter of urinary bladder .
 α_{1a} subtypes stimulation cause contraction and closure of the sphincters (**ppt urinary retention**)

1. Phenyl Ephrine

- It is relatively selective α_1 -agonist
- It is directly acting
- PK: not-catecholamine and thus not metabolized by COMT
- It has longer duration of action than other catecholamines

Clinical uses:

- As a mydriatic agent to examine the fundus of the eye
 - It acts on α_1 – receptors in the radial dilator pupillary muscle
- As a decongestant
 - Used as nasal drops to cause VC in the nasal blood vessels & relief congestion
- As a vasopressor agent in case of hypotension
 - α_1 stimulation causes VC leading to increase BP

Xylometazoline and oxymetazoline

- **Xylometazoline** and **oxymetazoline** are direct-acting α -agonists.
- **Use-** topical decongestants because of their ability to promote constriction of the nasal mucosa.

Adverse effect-

- oxymetazoline may cause hypotension (large doses)-may be due to α_2 agonist effect

α_2 - selective agonists

α_2 - selective agonists

Activate presynaptic α_2 receptors in the cardiovascular control center in the CNS => reduced sympathetic nervous system activity => blood pressure decrease

α_2 stimulation leads to decrease Norepinephrine release in adrenergic nerve terminals (presynaptic).

causing

– relaxation of smooth muscle

- In platelets.
 - Increase platelets aggregation

Clinical applications:

Hypertension

- Clonidine
- Guanfacine

Clonidine

- It is α_2 – selective agonist
- However, this is sympatholytic agent, used in treatment of hypertension
 - It acts centrally at presynaptic α_2 -adrenoceptor. This leads to decrease in NE release and to decrease in PVR.
 - **Note:** Although it is adrenergic agonist, clonidine acts as a central sympatholytic drug.
- Overdose stimulates peripheral **postsynaptic α_1 adrenoceptors** & cause **hypertension by vasoconstriction**

Clonidine

Use- treatment of systemic hypertension.

– Low dose Clonidine (50-100µg/dl) is used in migraine prophylaxis

Adverse effects:

- dry mouth
- Sedation, depression
- Sexual dysfunction
- Bradycardia
- Withdrawal syndrome (rebound hypertension) follows abrupt discontinuation of long-term therapy with clonidine in some hypertensive patients.

Methyldopa

- Centrally acting antihypertensive agent.
- metabolized to α -methylnorepinephrine in the brain
 - activate central α_2 receptors and lower blood pressure in a manner similar to that of clonidine.
- Safe and preferable anti-hypertensive agent during pregnancy

Site of β 1 – adrenoceptors & effects of their stimulation

- In the heart.
 - β 1 stimulation causes
 - In S.A node : increase HR (+ve chronotropic)
 - In Myocardium tissue : increase contractility (+ve inotropic)
 - In Conducting system : increase conduction velocity (+ve dromotropic)
- In the Juxtaglomerular Apparatus of the kidney.
 - β 1 stimulation cause increased renin release. Then causes increase in BP

Site of β 1 – adrenoceptors & effects of their stimulation

β 1 - selective agonists

Clinical applications:

- Dobutamine: synthetic analogue of dopamine
 - Strong inotropic effect with little chronotropic effect => increase in cardiac output without significant increase in heart rate
- Used in certain types of shock (very low blood pressure) and heart failure(short-term treatment of impaired cardiac function after cardiac surgery, MI etc.)

Site of β_2 -adrenoceptor & effects of their stimulation

- In the bronchial smooth muscle (very important clinically).
 - β_2 stimulation causes relaxation of smooth muscle (bronchodilatation)
- In the smooth muscle of blood vessels supplying the skeletal muscle.
 - β_2 stimulation causes relaxation of smooth muscle (Vasodilatation)
- In the smooth muscle of GIT wall.
 - β_2 stimulation cause relaxation of the wall leading to decreased peristalsis
- In the smooth muscle of the wall of urinary bladder.
 - β_2 stimulation causes relaxation of the wall (opposite to ACH)

Site of β_2 -adrenoceptor & effects of their stimulation

- In the smooth muscle of the uterus
 - β_2 stimulation causes **relaxation of the uterus**

So, β_2 agonists
- In the liver.
 - β_2 stimulation causes increased **Glycogenolysis & Gluconeogenesis**
- In the pancreas.
 - β_2 stimulation causes slight increase in insulin secretion

Then, what is the effect of β_2 stimulation on blood sugar?

β_2 - selective agonists

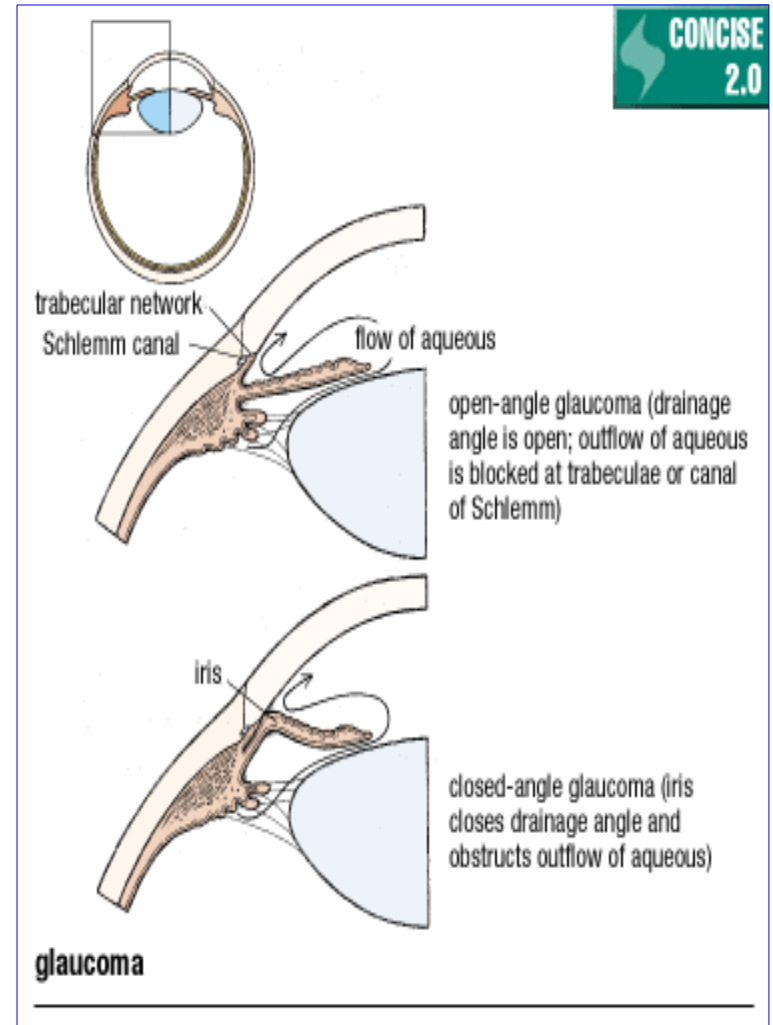
Asthma:

- β_2 - selective agonists target predominantly the respiratory system

β_2 – adrenoceptors...

- In ciliary muscle.
 - β_2 stimulation causes relaxation of the ciliary muscle leading to
 - Accommodation for far vision
 - Decrease outflow of aqueous humor via the canal of Schlemm
- In the ciliary epithelium
 - β_2 stimulation causes increased production of aqueous humor

Can we use b-adrenergic agonists for glaucoma?



β_2 -Selective Drugs

- widely used for the treatment of asthma.
- Short acting : Salbutamol, Metaproterenol, Terbutaline
 - Use- long-term treatment of *obstructive airway diseases, asthma*, and for *treatment of acute bronchospasm*.
- Long acting: Salmeterol, formeterol
 - *use -nocturnal asthma*.

β_2 -Selective Drugs

Ritodrine

- It is another β_2 – selective agonist
- It is used to **delay premature labour**
 - β_2 stimulation leads to relaxation of uterine smooth muscle leading to delay of labour.
 - This is done to ensure adequate maturation of fetus

Adverse effects of Beta-2 agonists

- *Skeletal muscle tremor*
 - Tolerance generally develops to this side effect
- *Restlessness, apprehension, and anxiety*
- *Tachycardia*
 - Stimulation of β_1 receptors OR secondary to β_2 receptor mediated vasodilation.
- Cardiac arrhythmia
- hyperglycemia

Indirectly Acting Adrenomimetic Drugs

1. Tyramine

- by-product of tyrosine metabolism in the body
- found in high concentrations in some foods such as cheese and fish.
- readily metabolized by MAO in the liver.
- has an indirect sympathomimetic action caused by the release of stored catecholamines.
 - has structural similarity to NE and E, and thus can be accumulated by *uptake* and *aminotransporter* into synaptic vesicles, displacing the neurotransmitters.

tyramine

- With MAO inhibitors, this effect of tyramine may be greatly intensified, leading to marked increases in blood pressure.
- Patients taking MAO inhibitors must be very careful to avoid tyramine-containing foods.

- indirectly acting sympathomimetic
- It is non-selective adrenergic agonist, non-catecholamine
 - Acts mainly, indirectly via **enhancing NE** release and **DA**.
 - blocks its reuptake into the cytoplasm of the nerve terminal.
 - has potent peripheral effects on α_1 , α_2 and β_1 receptors, but not β_2 receptors.
 - **Since it is non-catecholamine, it can be given orally**
 - **It is lipid-soluble enough to be absorbed from intestines and goes to all parts including CNS (This leads to CNS stimulation like Restlessness and Insomnia).**
 - $t_{1/2} = 45 - 60$ min (long duration of action)
 - It is metabolized in the Liver

Clinical use of Amphetamine-like drugs

- To suppress appetite
 - In very obese persons.
- In narcolepsy
 - Narcolepsy is irresistible attacks of sleep during the day in spite of enough sleep at night
- Decreased sense of fatigue
- Elevation of mood, with increased initiative, self-confidence
- Ability to concentrate
- Physical performance in athletes is improved
- Attention deficit-hyperactivity disorder /ADHD/

Amphetamines

Side effects

- The side effects are due to chronic use
- These include :
 - Tolerance
 - Dependence
 - Addiction
 - Psychosis
 - Hypertension

Cocaine

Cocaine is a local anesthetic with a peripheral sympathomimetic action that results from inhibition of transmitter reuptake at noradrenergic synapses. readily enters the central nervous system and produces an amphetamine-like effect that is shorter lasting and more intense. in CNS inhibit dopamine reuptake into neurons in the "pleasure centers" of the brain.

Mixed-acting Agent

Ephedrine

- Plant alkaloid, indirect sympathomimetic actions resembling adrenaline peripherally
- Is plant alkaloid, has both directly- and indirectly-mediated sympathomimetic effects.
 - stimulates **both** α and β receptors, and it causes release of NE.

Therapeutic use

- Bronchodilator (Theoephedrine tablet- 11mg ephedrine with 120mg theophylline in asthma)
- nasal decongestant
- Mydriatic
- heart block
- mucosal vasoconstriction

Adverse Effects of Ephedrine

- Symptoms of overdose are related primarily to cardiac and CNS effects.
 - Tachycardia, premature systoles, insomnia, nervousness, nausea, vomiting, and emotional disturbances may develop.

Contraindication

- cardiac disease
- Hypertension
- hyperthyroidism.

Clinical applications of Sympathomimetic drugs

- In [hypotension](#)
 - we use Phenyl Ephrine
- To reduce BF in certain organs
 - We use Adrenaline with local anesthetic for minor surgery in order to
 - Decrease bleeding
 - Prevent spread of local anesthetic into systemic circulation
 - Prolong the DOA of LA
- In [bronchial asthma](#)
- For [mydriasis](#)
- For [delaying of labor](#)

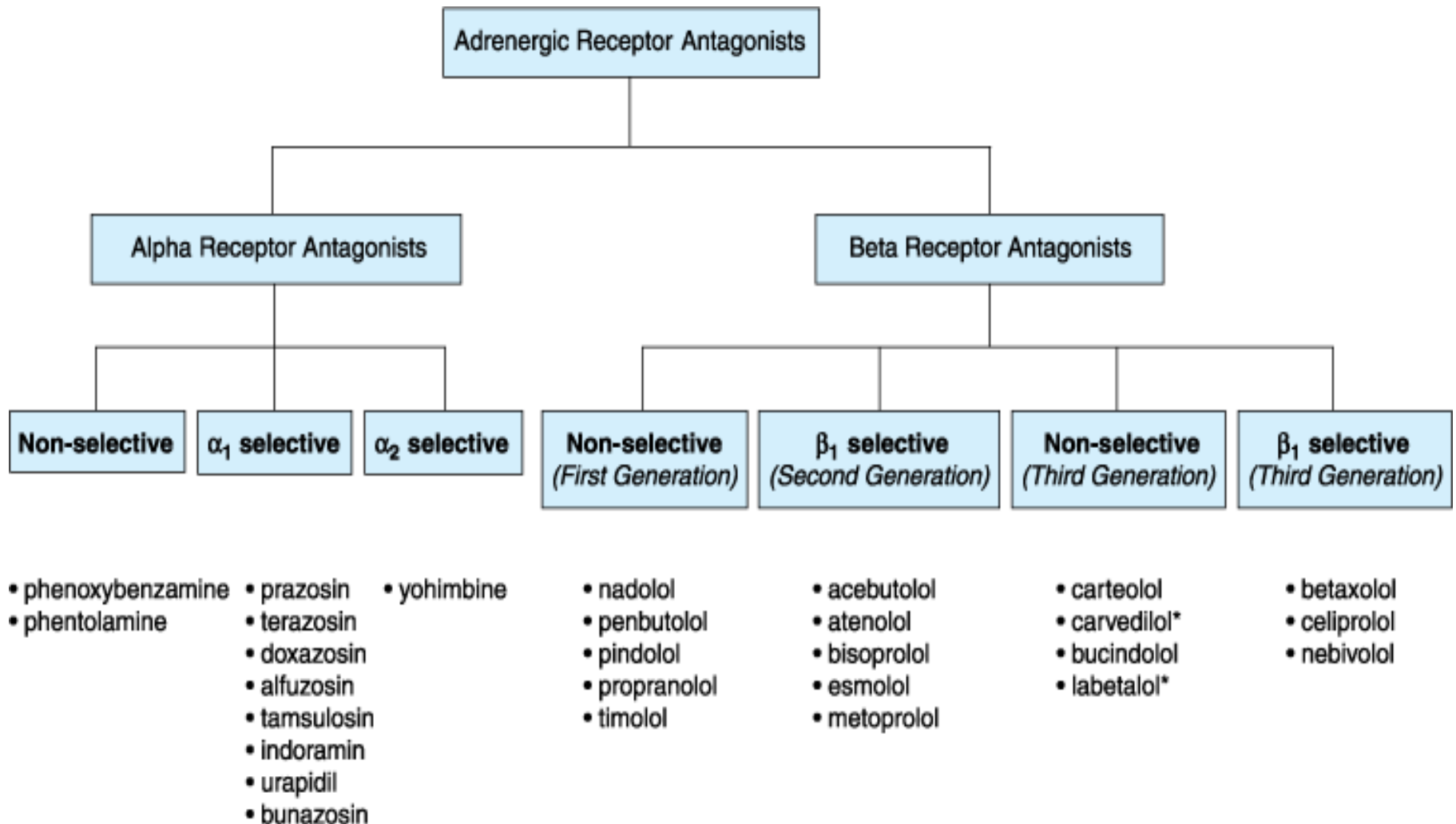
Side – effects of sympathomimetic drugs:

- On the CVS
 - Hypertension
 - Cardiac arrhythmia
 - Myocardial infarction
 - Increased severity of angina pectoris and of myocardial infarction
- On the eye
 - Increased I.O.P leading to Glaucoma
- On the CNS(Restlessness, anxiety)
- Tachycardia (β_2)
- Bradycardia(alpha 1)

Adrenergic receptor blockers (sympatholytics)

Adrenergic Blockers (Sympatholytics)

- Bind to adrenergic receptors, but inhibit or block stimulation of the sympathetic nervous system (SNS)



Alpha blockers

i. irreversible blockers

- in their interaction with these receptors; irreversible drugs do not dissociate.
 - **Phenoxybenzamine**

ii. reversible blockers

- Reversible antagonists dissociate from receptors

a. Non-selective α antagonists

- **Phentolamine, Tolazoline and Ergot derivatives**

b. Selective α -antagonists

- **α_1 Selective antagonists:**
 - prazosin, doxazosin, terazosin, and tamsulosin
- **α_2 selective antagonists:**
 - Yohimbine

Therapeutic Uses of α_1 - Adrenergic Receptor Blockers

1/Pheochromocytoma

- a tumor of the adrenal medulla or sympathetic ganglion cells.
- The tumor secretes catecholamines, especially NE and epinephrine.
- intermittent or sustained hypertension, headaches, palpitations, and increased sweating.
- useful in the preoperative management of patients with pheochromocytoma
- The tumor may be surgically removable, and it is essential to block α - and β -adrenoceptors before surgery is begun to avoid the effects of a sudden release of catecholamines when the tumour is disturbed.
- A combination of phenoxybenzamine and **atenolol** is effective for this purpose.

Therapeutic Uses of α_1 - Adrenergic Receptor Blockers

2/Hypertension

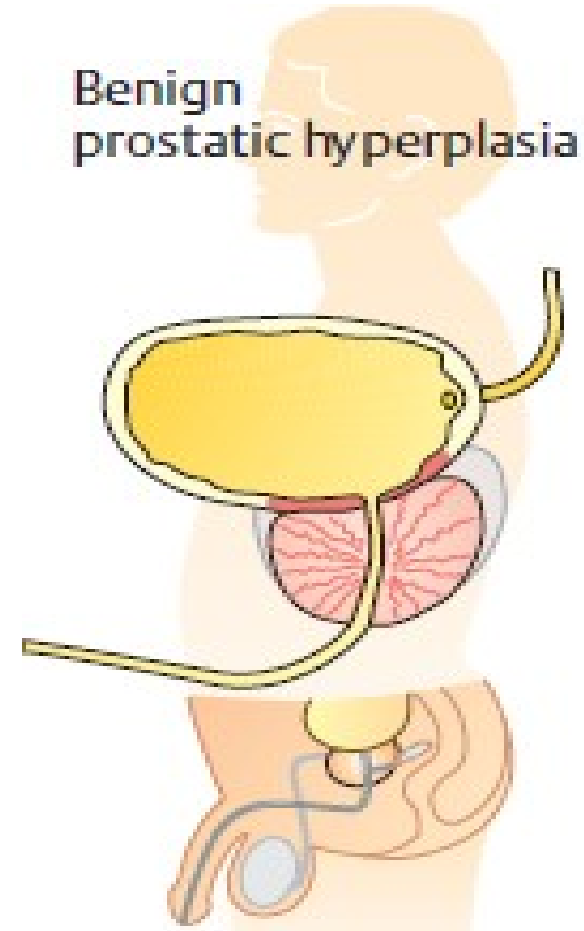
3/ treatment of peripheral vasospastic disease ie Raynauds disease to improve perfusion

4/ Treatment of local excess concentration of a vasoconstrictor in order to prevent necrosis.

Therapeutic uses of α_1 - adrenergic receptor blockers...

5/ Urinary Obstruction: BPH

- common in elderly men: weak stream, urinary frequency, and nocturia.
- surgical treatments
- **drug therapy : Prazosin, doxazosin, terazosin** tamsulosin/Alpha1A/
- improving urine flow
- partial reversal of smooth muscle contraction in the enlarged prostate and in the bladder base.
- Prazosin, terazosin and doxazosin used in the treatment of BPH
 - Due to induction of apoptosis in prostate smooth muscle cells.
 - may lessen the symptoms associated with chronic BPH by limiting cell proliferation.



α_1 - adrenergic receptor blockers; Adverse effects

- Postural hypotension
- Tachycardia
- Sedation
- Nasal stuffiness
- Meiosis
- inhibits ejaculation

Pharmacologic effects of α_2 Blockers

- e.g. yohimbine
- increase sympathetic outflow
- potentiate the release of norepinephrine from nerve endings, leading to activation of α_1 and β_1 receptors in the heart and peripheral vasculature with a consequent rise in blood pressure.
- It has no established clinical role.
- Theoretically, it could be useful in autonomic insufficiency by promoting neurotransmitter release through blockade of presynaptic α_2 receptors.
- Yohimbine can abruptly reverse the antihypertensive effects of an α_2 -adrenoceptor agonist such as clonidine

Non-selective α antagonists

- **Phenoxybenzamine**
 - not specific irreversible blocker of α -adrenoceptors
 - antagonises the actions of acetylcholine, histamine and 5-HT.
- The major use of phenoxybenzamine is in the treatment of pheochromocytoma.

Phentolamine

- potent competitive antagonist at both α_1 and α_2 receptors.
- Phentolamine causes a reduction in peripheral resistance through blockade of α_1 receptors.
- The cardiac stimulation induced by phentolamine is due to sympathetic stimulation of the heart resulting from baroreflex mechanisms and antagonism of presynaptic α_2 receptors may lead to enhanced release of norepinephrine from sympathetic nerves.

Phentolamine is used for Erectile Dysfunction

- A combination of **phentolamine** with the nonspecific smooth muscle relaxant papaverine, when injected directly into the penis, may cause erections in men with sexual dysfunction.

β - Adrenoceptor Antagonists

- B-blocking drugs occupy B-receptors and competitively reduce receptor occupancy by catecholamines and other B- agonists.

β -Blockers: Examples

- Carvedilol
 - labetalol
- } non-selective α/β blocker

- Atenolol
 - Metoprolol
- } selective β_1 blocker

- Acebutolol
 - Esmolol
 - Sotalol
 -
- } non- selective β blocker
- propranolol

Beta Blockers

Pharmacological Actions

CVS:

a. Heart (β_1):

- ↓ myocardial contraction
- ↓ HR
- ↓ AV-conduction & automaticity

b. BV to Sk. Muscle (β_2):

Vasoconstriction, muscle pain & fatigue (↑ with non- selective)

c. JG cells (β_1):

↓ renin release, ↓ aldosterone , ↓ angiotensin :
» ↓ Na & water retention,

Pharmacological Actions...

- EYE:
 - ↓ IOP
- CNS/Neurological:
 - Sedation (propranolol, carvedilol)
- Respiratory:
 - Broncho-constriction,
- Metabolic:
 - ↓ gluconeogenesis ↓ glycogenolysis and lipolysis

Clinical applications of β -Blockers

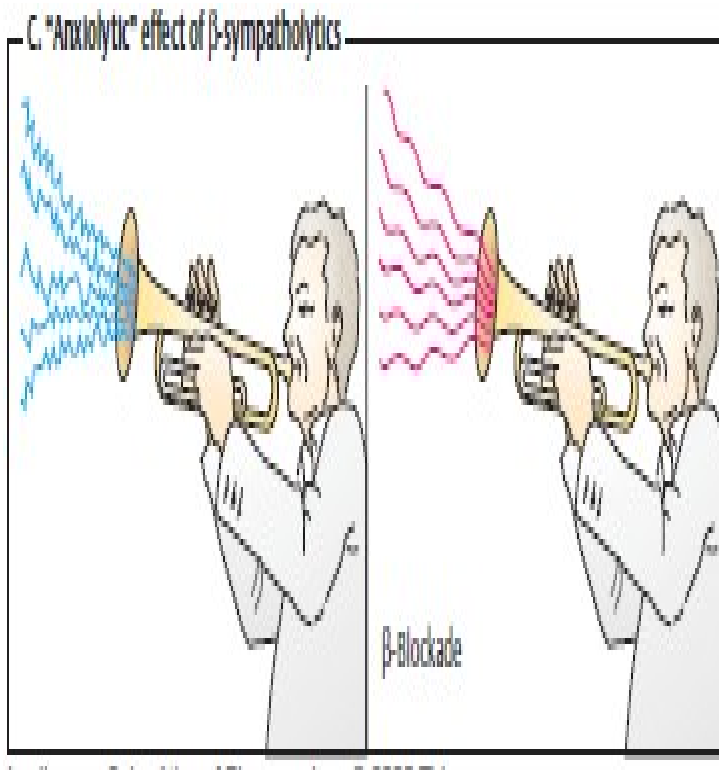
- **Antiangina:** Decreases demand for myocardial oxygen
- **Cardioprotective:** Inhibits stimulation from circulating catecholamines(pheochromocytoma)
- **Class II antiarrhythmics**
- **Antihypertensive**
- Some are used to treat **heart failure** (early stage to prevent cardiac remodeling)

C. Glaucoma: (timolol)

- diminish IOP in glaucoma by decreasing the secretion of aqueous humor by the ciliary body.

d. Hyperthyroidism: blunting the widespread sympathetic stimulation that occurs in hyperthyroidism

Clinical applications...



- -Migraine prophylaxis:
blockade of
catecholamine-induced
vasodilation in the brain
vasculature
- **tremor**
- **performance Anxiety /
stage fright**

Beta-Blockers - Adverse Effects

- B1 –blockade /cardiac/: -bradycardia; hypotension
- B2 blocked :Bronchoconstriction; muscle fatigue
- PVD
- Decreased blood flow to vital organ
- Metabolic: - Increase hypoglycemic effect of insulin(IDDM)

NB- can also mask symptoms of hypoglycemia

- CNS; depression, nightmares
- Sudden withdrawal : angina pectoris ,sudden death
 - tapering the dose of the B- blocker for several weeks before discontinuation.

Cardiovascular Pharmacology

By: Fantu.K (B.pharm, Msc, assit. Prof.)

Anti-hypertensive Agents

Objectives

- Describe types and classification of hypertension
- Describe **Causes of Secondary Hypertension**
- Discuss the different treatment approaches of **Hypertension**
- Describe the main **humoral mechanism** involved in the development of essential hypertension.

Introduction

- Hypertension is a persistently elevated arterial blood pressure (BP).
- Arterial BP is the pressure in the arterial wall measured in millimeters of mercury (mm Hg).
- The two typical arterial BP values are *systolic BP (SBP)* and *diastolic BP (DBP)*.
- *SBP is achieved during* cardiac contraction and represents the peak value.
- DBP is achieved after contraction when the cardiac chambers are filling, and represents the nadir value.

Etiology

- In most patients, hypertension results from an unknown pathophysiologic etiology (*essential or primary hypertension*).
- More than 90% of individuals with hypertension have essential hypertension.
- *This form of HTN* cannot be cured, but it can be controlled.
- Hypertension is a heterogeneous disorder that may result either from a specific cause (**secondary hypertension**)

Etiology

- Secondary hypertension accounts for fewer than 10% of cases, and most of these are caused by **chronic kidney disease or renovascular disease.**
- If the cause can be identified, hypertension in these patients has the potential to be cured.

Causes of Secondary Hypertension

Diseases

- Chronic kidney disease
- Cushing's syndrome
- Pheochromocytoma
- Primary aldosteronism
- Renovascular disease
- Prescription drugs e.g. Amphetamines, Decongestants

Classification of Blood Pressure in Adults

(Age ≥18 Years)

Classification	Systolic Blood Pressure (mm Hg)	Diastolic Blood Pressure (mm Hg)
Normal	<120 and	<80
Prehypertension	<i>120–139</i>	<i>80–89</i>
Isolated systolic hypertension	≥ 140	< 90
Stage 1 hypertension	140–159	90–99
Stage 2 hypertension	≥160	≥100
Hypertensive crisis	>180	>120

Renin-angiotensin-aldosterone system

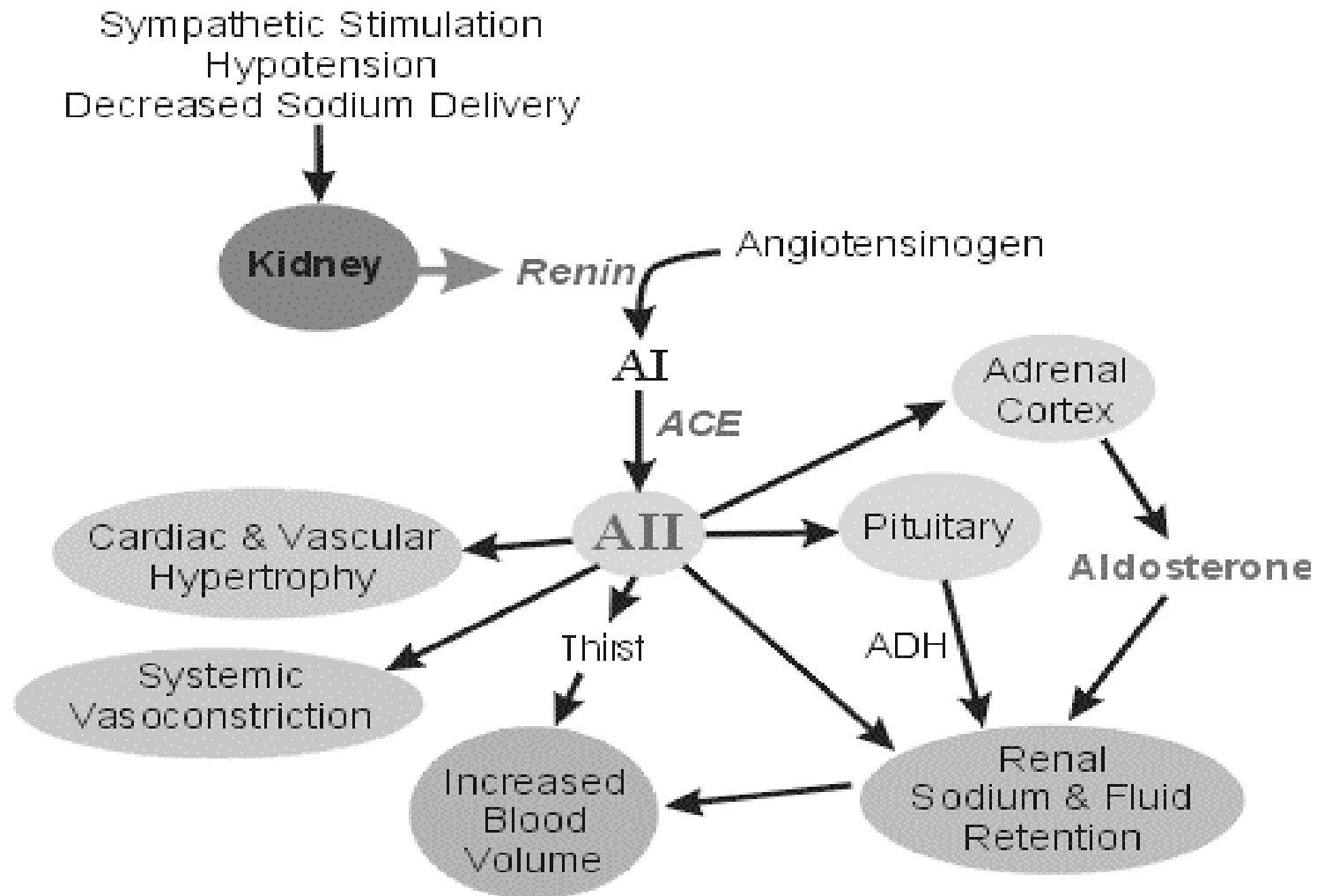
- Important role in regulating blood volume, arterial pressure, and cardiac and vascular function.
- Most important site for renin release is the kidney: sympathetic stimulation (acting via β_1 -adrenoceptors), renal artery hypotension (e.g. stenosis), and decreased sodium delivery to the distal tubules stimulate the release of renin by the kidney.
- Renin acts upon a circulating substrate, angiotensinogen which undergoes proteolytic cleavage to form the angiotensin I (AT I).
- Vascular endothelium, particularly in the lungs, contains angiotensin converting enzyme (ACE), which cleaves off two amino acids to form the angiotensin II (AT II).

Renin-angiotensin –aldosteron system

Angiotensin II

- Constricts vessels thereby increasing vascular resistance and arterial pressure
- Stimulates the adrenal cortex to release aldosterone, which acts upon the kidneys to increase sodium and fluid retention
- Stimulates the release of vasopressin (ADH) from the pituitary which acts upon the kidneys to increase fluid retention
- Facilitates NE release and inhibits reuptake from nerve endings, thereby enhancing sympathetic adrenergic function
- Stimulates cardiac and vascular hypertrophy

Renin-angiotensin system



Risk factors

- Age (≥ 55 years for men and 65 years for women)
- Diabetes mellitus
- Dyslipidemia (elevated low-density lipoprotein-cholesterol, total cholesterol, and/or triglycerides; low high densitylipoprotein-cholesterol)
- Family history of premature CV disease
- Obesity (body mass index ≥ 30 kg/m²)
- Physical inactivity
- Tobacco use

- There are 2 treatment approaches:

1. Non pharmacological

2. Pharmacological

Life style modification

1) Weight reduction

.For every 10kg lost, SBP ↓ by 5-20mmHg

.Even small amount has impressive consequences

2) Exercise

.Even without weight loss ,it lowers incidence of HTN & protects against CV disease

.30 min of aerobic activity >3 X /week

SBP ↓ by 4 – 9mmHg

Life style modification

3) Dietary changes

Sodium- No more than 100mmol/d

2.4g Na=6gm of NaCl

- ↓SBP by 2-8mmHg

- Diet rich in vegetables & fruits

.Low fat dairy product with less saturated & total fat

4) Moderation of alcohol

5) Avoidance of tobacco

.sympathetic outflow ↑es with each cigarette—
arterial stiffness

Classification of antihypertensive agents

- The categories include the following:
 - (1) **Diuretics**, which lower blood pressure by depleting the body of sodium and reducing blood volume.
 - (2) **Sympathoplegic agents**, which lower blood pressure by reducing peripheral vascular resistance, inhibiting cardiac function
 - (3) **Direct vasodilators**, which reduce pressure by relaxing vascular smooth muscle.
 - (4) **Agents that block production or action of angiotensin** and thereby reduce peripheral vascular resistance and (potentially) blood volume.

Antihypertensive Agents...

- Diuretics
- ACE inhibitors
- ARBs
- Calcium channel blockers
- β -Blockers
- B + α Adrenergic blockers
- α Adrenergic blockers
- Central sympatholytics
- Vasodilators
- Ganglionic blockers

ACE inhibitors

- The ACE inhibitors are one of the first choice drugs in all grades of essential as well as renovascular hypertension
- Benazepril
- Captopril
- Enalapril
- Fosinopril
- Lisinopril
- Moexipril
- Perindopril
- Quinapril
- Ramipril
- Trandolapril

Mechanism of action

- ACE facilitates production of angiotensin II, which has a major role in regulating arterial blood pressure.
- ACE principal location is in endothelial cells.
- ACE inhibitors block the conversion of angiotensin I to angiotensin II, a potent vasoconstrictor and stimulator of aldosterone secretion.
- ACE inhibitors also block the degradation of bradykinin and stimulate the synthesis of other vasodilating substances including prostaglandin E₂ and prostacyclin.

Adverse effects

- Common side effects of ACE inhibitors include
- dry cough,
- Angioedema
- Dizziness and low blood pressure.
- fatigue,
- headache.
- can cause increased potassium levels or kidney problems.

Angiotensin II Receptor Blockers

- Candesartan
- Eprosartan
- Irbesartan
- Losartan
- Olmesartan
- Telmisartan
- Valsartan
- May cause hyperkalemia in patients with chronic kidney disease or in those receiving potassium-sparing diuretics(aldosterone antagonists) ACE inhibitors, or direct renin inhibitor
- Do not cause a dry cough and angioedema like ACE inhibitors ;

Mechanism of action

- Angiotensin II is generated by the renin-angiotensin pathway (which involves ACE) and an alternative pathway that uses other enzymes such as chymases.
- ACE inhibitors block only the renin-angiotensin pathway, whereas ARBs antagonize angiotensin II generated by either pathway.
- The ARBs directly block the angiotensin type 1 (AT₁) receptor

Angiotensin II Receptor Blockers...

- Unlike ACE inhibitors, ARBs do not block the breakdown of bradykinin.
- While this accounts for the lack of cough and angioedema as a side effect, there may be negative consequences because some of the antihypertensive effect of ACE inhibitors may be due to increased levels of bradykinin.

Adverse effects

- Cough is very uncommon.
- Renal insufficiency,
- Hyperkalemia,
- Orthostatic hypotension.
- Angioedema is less likely to occur than with ACE inhibitors,
- ARBs should not be used in pregnancy.

Calcium channels blockers

- The dihydropyridines (DHPs) are the most potent Ca^{2+} channel blockers.

Mechanism of action of calcium channel blockers

- The common property of all CCBs is to inhibit Ca^{2+} mediated slow channel component of action potential (AP) in smooth/cardiac muscle cell.
- The two most important actions of CCBs are:
 - Smooth muscle (especially vascular) relaxation.
 - Negative chronotropic, inotropic and dromotropic action on heart.

Calcium channels blockers...

Dihydropyridines:

- Amlodipine
- Nicardipine SR
- Nifedipine long-acting
- Nisoldipine

Adverse effects

- More reflex sympathetic discharge (tachycardia), dizziness, headache, flushing, and peripheral edema; have additional benefits in Raynaud's syndrome

Calcium channels blockers...

Non-dihydropyridines:

- Diltiazem
- Verapamil

Adverse effects

- Patients receiving β -blocking drugs are more sensitive to the cardiodepressant effects of calcium channel blockers.
- dizziness, headache, flushing, and peripheral edema.
- Constipation is particularly common with verapamil

Clinical uses

- Hypertension
- Cardiac arrhythmias
- Hypertrophic cardiomyopathy
- Nifedipine is an alternative drug for premature labour
- Myocardial infarction
- Angina pectoris

Pharmacologic Effects of Calcium Channel Blockers

Effect	Verapamil	Diltiazem	Dihydropyridines
Peripheral Vasodilation	↑	↑	↑↑
Heart Rate	↓↓	↓	↑
Cardiac Contractility	↓↓	↓	0 / ↓
SA / AV Nodal Conduction	↓	↓	0
Coronary Blood Flow	↑	↑	↑↑

Alpha₁-adrenergic antagonists

Mechanism:

- Blocking alpha-1 receptors: doxazosin, prazosin, and terazosin

Therapeutic Uses of Alpha Adrenergic Receptor Blockers

1/Pheochromocytoma

- useful in the preoperative management of patients with pheochromocytoma

2/Hypertension

3/ treatment of peripheral vasospastic disease ie Raynauds disease to improve perfusion

4/ Treatment of local excess concentration of a vasoconstrictor in order to prevent necrosis.

5/ Urinary Obstruction: BPH

- improving urine flow
- Reversal of smooth muscle contraction in the bladder base(internal sphincter of urinary bladder)

Alpha blockers; Adverse effects

- Postural hypotension
- Tachycardia
- Sedation
- Nasal stuffiness
- inhibits ejaculation

α_2 - selective agonists

Activate presynaptic α_2 receptors in the cardiovascular control center in the CNS => reduced sympathetic nervous system activity => blood pressure decrease

α_2 stimulation leads to decrease Norepinephrine release in adrenergic nerve terminals (presynaptic).

Clinical applications:

Hypertension

- Clonidine
- Guanfacine
- Methyldopa

Clonidine

- It is α_2 – selective agonist
- However, this is sympatholytic agent, used in treatment of hypertension
 - It acts centrally at presynaptic α_2 -adrenoceptor. This leads to decrease in NE release and to decrease in PVR.
- Overdose stimulates peripheral **postsynaptic α_1 adrenoceptors** & cause **hypertension by vasoconstriction**

Clonidine

Adverse effects

- Sedation, depression
- Sexual dysfunction
- Bradycardia
- Withdrawal syndrome (rebound hypertension) follows abrupt discontinuation of long-term therapy with clonidine in some hypertensive patients.

Methyldopa

- Centrally acting antihypertensive agent.
- metabolized to α -methylnorepinephrine in the brain
 - activate central α_2 receptors and lower blood pressure in a manner similar to that of clonidine.
- Safe and preferable anti-hypertensive agent during pregnancy

- The primary indication: mild and moderate hypertension
—that has not responded adequately to treatment with a diuretic or α_1 -blockers.
- For severely hypertensive patients, *clonidine has been used in combination with a diuretic, a vasodilator, and alpha1-blocker.*

β- blockers

- Selective beta 1 blockers decrease cardiac output through negative **chronotropic and inotropic** effects on the heart and inhibition of renin release from the kidney.

Cardioselective beta blockers

- Atenolol
- Betaxolol
- Bisoprolol
- Metoprolol tartrate
- Metoprolol succinate
- They are less likely to provoke bronchospasm and vasoconstriction and may be safer than nonselective β blockers in patients with asthma, chronic obstructive pulmonary disease (COPD), diabetes, and peripheral arterial disease.

β-Blockers

Nonselective:

- Nadolol
- Propranolol
- Timolol (Blocadren)
- Abrupt discontinuation may cause rebound hypertension;
- Inhibit β₁- and β₂-receptors at all doses(can exacerbate asthma...)
- Have additional benefits in patients with tremor, migraine headache, thyrotoxicosis

Clinical uses

- hypertension
- Angina pectoris
- Myocardial Infarction
- Heart Failure

Adverse effects

- Bradycardia
- Atrioventricular (AV) conduction abnormalities
- Acute heart failure.
- Acute exacerbations of bronchospasm in patients with asthma or COPD.
- Cold extremities
- Raynaud's phenomenon because of decreased peripheral blood flow.

Vasodilators

- **Vasodilators** produce a direct relaxation of vascular smooth muscle and thereby result in vasodilation.
- The vasodilators decrease total peripheral resistance and thus correct the hemodynamic abnormality that is responsible for the elevated blood pressure in primary hypertension.
- In addition, because they act directly on vascular smooth muscle, the vasodilators are *effective in lowering blood pressure, regardless of the etiology of the hypertension*

Vasodilators: **Classes**

- **Arterial vasodilators**
 - Minoxidil, Diazoxide: **Potassium Channel Openers**
 - *Hydralazine*

Hydralazine

- The vasodilation depends in part on the presence of an intact blood vessel endothelium.
 - This implies that hydralazine causes the release of nitric oxide, which acts on the vascular smooth muscle to cause relaxation.
- **Arterial and venous dilators:**
 - Sodium nitroprusside: Releases NO

Clinical Uses

- Hydralazine is generally reserved for moderately hypertensive ambulatory patients
 - whose blood pressure is not well controlled either by diuretics or
 - by drugs that interfere with the sympathetic nervous system.
- It is almost always administered in combination with
 - a diuretic and beta-blocker

Adverse Effects

headache, flushing, nasal congestion and tachycardia.

Minoxidil

- Minoxidil (*Loniten*) is an orally effective vasodilator.
- *It* is more potent and longer acting than hydralazine
- It depends on *in vivo* metabolism by hepatic enzymes to produce an active metabolite, ***minoxidil sulfate***.
- Minoxidil sulfate activates potassium channels,
 - resulting in hyperpolarization of vascular smooth muscle and relaxation of the blood vessel.

Clinical Uses

- Severe hypertension that may be life threatening and
- hypertension that is resistant to milder forms of therapy.

Adverse Effects

- include headache, nasal congestion and tachycardia.

Diazoxide

- Diazoxide *is chemically similar to the thiazide* diuretics.
 - But devoid of diuretic activity.
- Diazoxide is a very potent vasodilator and is available only for intravenous use in the treatment of hypertensive emergencies.
- The mechanism by which diazoxide relaxes vascular smooth muscle is related to its ability to
 - activate potassium channels and produce a hyperpolarization of the cell membrane

Clinical Uses

- Diazoxide is administered intravenously for the treatment of *hypertensive emergencies*
- particularly
 - malignant hypertension and
 - eclampsia**.
- It is effective in 75 to 85% of the patients to whom it is administered and rarely reduces blood pressure below the normotensive range.

Adverse Effects

- The chief concern is the side effects associated with the increased workload on the heart,
 - which may precipitate myocardial ischemia and
 - Na and water retention.
- may stop labor, because it relaxes uterine smooth muscle.

Arterial and venous vasodilators: *Sodium Nitroprusside*

- Sodium nitroprusside *is a potent directly acting* vasodilator
 - capable of reducing blood pressure in all patients, regardless of the cause of hypertension.
- It is used only by the intravenous route
 - for the treatment of *hypertensive emergencies*.
- The actions of the drug are similar to those of the nitrites and nitrates that are used as antianginal agents.
 - depends on the **intracellular production of cGMP**.

Clinical Uses

- Sodium nitroprusside is used in the management of hypertensive crisis.
- Because the drug reduces preload and afterload,
 - it improves ventricular performance and in fact is sometimes used in patients with refractory heart failure, even in the absence of hypertension.

Adverse Effects

- nausea, vomiting, and headache

Ganglionic blocking agents

- The basis for these agents **lies in their ability to block transmission through autonomic ganglia.**
 - results in a decrease in impulses passing down the *postganglionic sympathetic* (and parasympathetic) nerves, decreases vascular tone, cardiac output, and BP.
- The ganglionic blocking agents are extremely potent antihypertensive agents and
- can reduce BP regardless of the extent of hypertension.

Ganglionic blocking agents...

Unfortunately, **the ganglionic blocking agents are rarely used.**

- They produces numerous untoward severe responses,
 - including marked postural hypotension, blurred vision, and dryness of mouth, constipation, paralytic ileus, urinary retention, and impotence.
- Certain intravenous preparations,
 - such as the short-acting agent **trimethaphan camsylate**,
 - *are used occasionally for hypertensive emergencies and*
 - in surgical procedures in which hypotension is desirable to reduce the possibility of hemorrhage.

Drugs used to treat hypotension

- Hypotension is BP below 100/60 mm Hg in women and less than 110/70 mm Hg in men.

Causes of hypotension

Diseases:

- Sever hemorrhage
- Heart diseases(Cardiac insufficiency and arrhythmias,)

Drugs

- Antihypertensive drugs
- Diuretics
- Anti-antianginal drugs
- Vasodilators
- Anti-arrhythmic drugs

Drugs used to treat hypotension.....

Sympathomimetic agents

- Used in hypotensive emergency to preserve cerebral and coronary blood flow.

Direct acting alpha agonists

Norepinephrine (Noradrenaline) is an agonist at

- both α 1($\uparrow$ peripheral resistance), α 2, & β 1 receptors.
 $\uparrow$ CO = $\uparrow$ BP.

Phenylephrine pure α 1 agonist.

- longer duration of action than the catecholamines.
- $\uparrow$ peripheral resistance = $\uparrow$ BP.
- **Methoxamine** α 1 -receptor agonist. $\uparrow$ peripheral resistance = $\uparrow$ BP.
- Prolonged duration of action

Shock

- Acute cardiovascular syndrome that results in a critical ↓ in perfusion of vital tissues & a wide range of systemic effects.
- Usually associated with hypotension, oliguria & metabolic acidosis.
- If untreated, it progresses to a refractory deteriorating state & death.

Mechanisms responsible for shock :

- Hypovolemia, cardiac insufficiency, & altered vascular resistance.
- Volume replacement & treatment of the underlying disease are the mainstays of the treatment of shock.

Shock

- **Cardiogenic shock and acute heart failure**, usually due to massive myocardial infarction.
- Dopamine or dobutamine : positive inotropic drugs
- Dopamine acts on
 - β_1 receptors (cardiac stimulation)
 - Dopaminergic receptors (renal vasodilation)
 - α_1 receptors (vasoconstriction)
- Dobutamine acts on β_1 (cardiac stimulation)

Shock

Anaphylactic shock :

Type I hypersensitivity reaction that affects both the respiratory & the cardiovascular systems causing

- bronchospasm, angioedema, and severe hypotension
- Inj. Adrenaline, 0.3–0.5 mg IM
- Activates α , β_1 , and β_2 receptors, all of which may be important in reversing the pathophysiologic processes underlying anaphylaxis
- Glucocorticoids and antihistamines may be useful as secondary therapy in anaphylaxis.

Pharmacology of Heart Failure

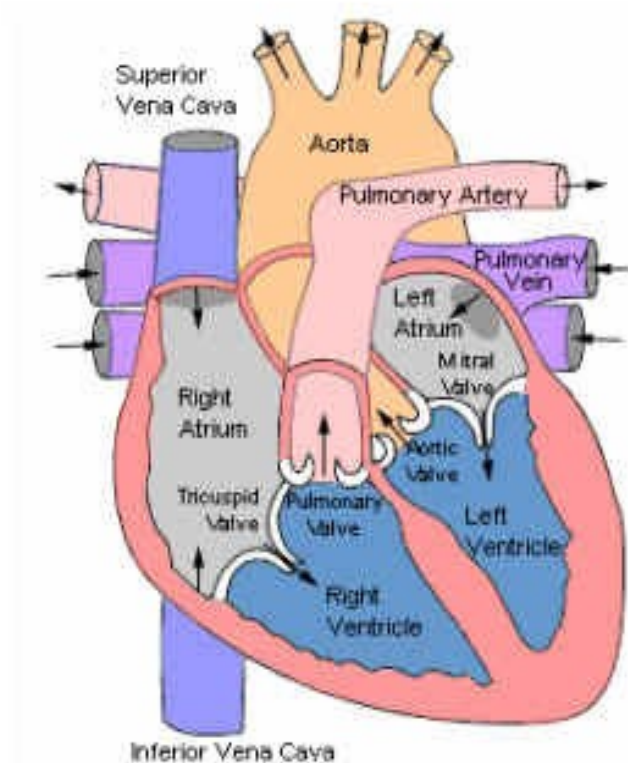
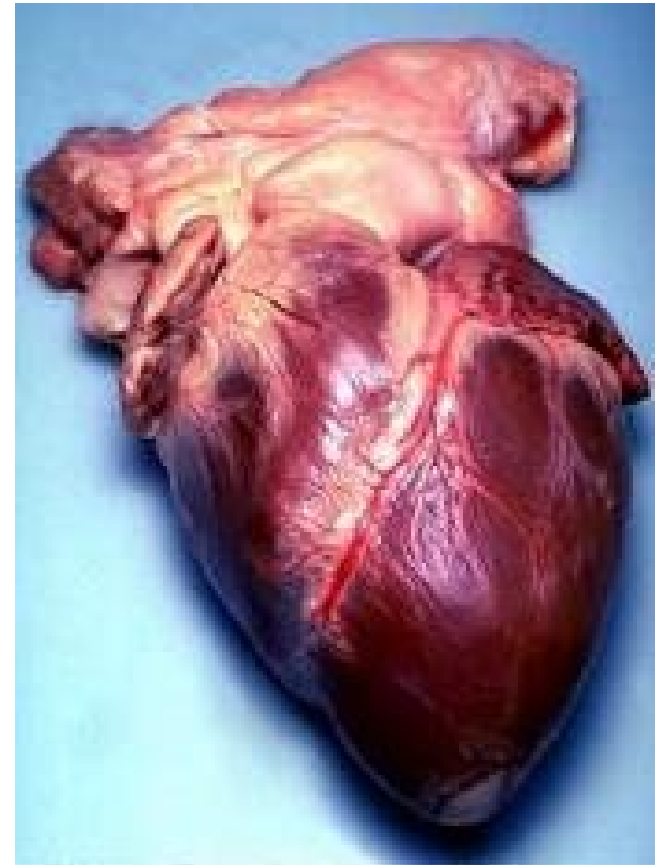


Figure 1: Diagram of the human heart showing the left and right ventricles along with the each of the four valves [HeartCenterOnline, 2002]



Objectives

- Describe etiology & Causes of Heart Failure
- Discuss the Basic Pharmacology Of Drugs Used In Heart Failure

Definition :HF

- Inability of the heart to pump an adequate amount of blood to the body's needs
- **CONGESTIVE HEART FAILURE** – refers to the state in which abnormal circulatory congestion exists as result of heart failure



Pathophysiology of HF

- Pump fails → decreased stroke volume /CO.
- Compensatory mechanisms kick in to increase CO
- ↓ renal perfusion interpreted by juxtaglomerular apparatus as hypovolemia. Thus: Kidneys release renin, which stimulates conversion of angiotensinogen → angiotensin I, angiotensin I to angiotensin II which causes:
 - SNS stimulation → release of epinephrine /norepinephrine
 - Increase HR
 - Increase contractility
 - Peripheral vasoconstriction (increases afterload)

Pathophysiology of HF...

- Myocardial hypertrophy: walls of heart thicken to provide more muscle mass → stronger contractions
- Aldosterone release → Na and water retention (via ADH secretion)
- Peripheral vasoconstriction
 - Helps shunt blood away from non-essential organs

Pathophysiology of HF...

- Compensatory mechanisms may restore CO to near normal. But, if excessive the compensatory mechanisms can worsen heart failure because:
 - Vasoconstriction:
 - $\uparrow$ the resistance against which heart has to pump (i.e., $\uparrow$ afterload), and may therefore $\downarrow$ CO
 - Na and water retention:
 - $\uparrow$ fluid volume, which $\uparrow$ preload. If too much “stretch” (due to too much fluid) $\rightarrow \downarrow$ strength of contraction and $\downarrow$ CO
 - Excessive tachycardia $\rightarrow \downarrow$ diastolic filling time $\rightarrow \downarrow$ ventricular filling $\rightarrow \downarrow$ SV and CO

Risk factors of HF

- CAD
- Age
- HTN
- Obesity
- Cigarette smoking
- Diabetes mellitus
- High cholesterol

Etiology of HF

- The most common causes of heart failure are coronary artery disease, high blood pressure, and diabetes.
- HF results from any disorder which prevents:
 - The heart from contracting (Systolic Dysfunction)
 - and/or**
 - The ventricles from relaxing and filling with blood (Diastolic Dysfunction)
- Systolic dysfunction is more prevalent
- Systolic and diastolic dysfunction may coexist

- Systolic Dysfunction
 - Reduction in muscle mass e.g. MI
 - Leads to systolic contractile dysfunction
 - Ventricular hypertrophy
 - pressure overload
 - volume overload
- Diastolic Dysfunction
 - Myocardial infarction may slow ventricular relaxation and increase ventricular stiffness
 - Ventricular hypertrophy

Pathophysiology of Cardiac Performance

Factor	Mechanism	Therapeutic Strategy
1. Preload (work or stress the heart faces at the end of diastole)	increased blood volume and increased venous tone--->atrial filling pressure	-salt restriction -diuretic therapy -venodilator drugs
2. Afterload (resistance against which the heart must pump)	increased sympathetic stimulation & activation of renin-angiotensin system ---> vascular resistance ---> increased BP	- arteriolar vasodilators -decreased angiotensin II (ACE inhibitors)
3. Contractility	decreased myocardial contractility ---> decreased CO	-inotropic drugs (cardiac glycosides)
4. Heart Rate	decreased contractility and decreased stroke volume	Increase HR (via activation of beta adrenoceptors)

Due to excess fluid accumulation

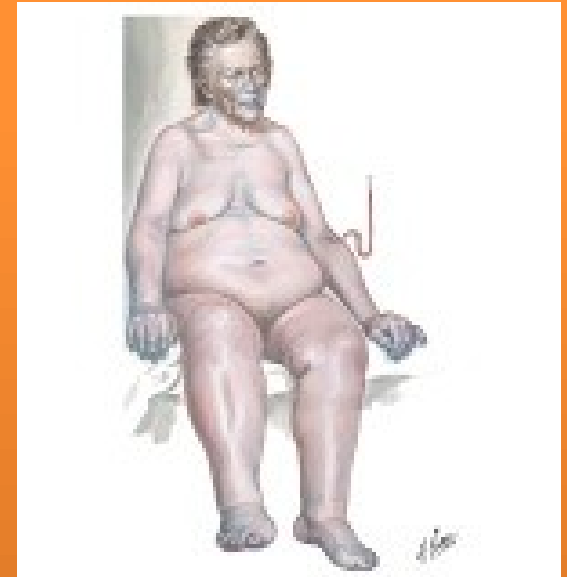
- Dyspnea
- Orthopnea
- Tachypnea
- Peripheral edema and ascites
- Cough

Due to reduction in cardiac output:

- Fatigue
- Exercise intolerance

Signs:

- Pulmonary edema
- Cardiomegaly



Treatment

❑ Non-pharmacologic therapy:

- Exercise training: for stable HF patients increased exercise capacity, decreased hospitalization rate, increased quality of life, decreased symptoms.
- Weight loss in obese patients
- Dietary Na restriction
- Fluid and free water restriction especially if hyponatremic
- Minimize medications known to have effects on heart failure (negative inotropes, NSAIDs, over-the-counter stimulants)
- Oxygen
- Fluid removal (dialysis)

ACE inhibitors for HF

- Decrease morbidity & the combined risk of death and hospitalizations by
 - ↓ long-term remodeling of the heart and vessels.
 - Many trials have documented reductions in mortality of 25%
 - ↓ peripheral resistance (↓ afterload)
 - ↓ reduce salt and water retention by ↓ aldosterone secretion (↓ preload).

ACE inhibitors for HF...

- ACE Inhibitors approved for the treatment of HF
 - captopril: Capoten[®]
 - enalapril: Vasotec[®]
 - lisinopril: Zestril[®], Prinivil[®]
 - quinapril: Accupril[®]
 - fosinopril: Monopril[®]
 - ramipril: Altace[®]
 - randolapril: Mavik[®]

Actions of ACEI on the heart

- **ACE** inhibitors decrease vascular resistance, venous tone, and blood pressure.
- ACE inhibitors also blunt the usual angiotensin II mediated increase in epinephrine and aldosterone seen in HF.
- The use of ACE inhibitors in the treatment of HF has significantly decreased both morbidity and mortality
- **Note:** Reduction in mortality is due primarily to a decrease in deaths caused by progressive HF.

Angiotensin receptor blockers

- large clinical trials suggest that angiotensin receptor blockers are best reserved for patients who cannot tolerate ACE inhibitors
- Losartan , telmisartan
- Block AT1 receptor on the heart, peripheral vasculature and kidney
- Produce beneficial effects similar to those of ACE inhibitors. Potential role of ARBs as adjunct therapy

Beta Blockers for HF

- Bisoprolol, carvedilol, metoprolol, and nebivolol showed a reduction in mortality in patients with stable severe heart failure
- A full understanding of the beneficial action of β blockade is lacking, but suggested mechanisms include attenuation of the adverse effects of high concentrations of catecholamines & up-regulation of β receptors
- Useful even though they have negative inotropic action
- Symptomatic improvement may occur after several months.

Diuretics for HF

- Diuretics, especially furosemide, are drugs of choice in heart failure
- Almost all symptomatic Patients treated with a diuretic
- High ceiling diuretics (loop diuretics) preferred
 - Low dose therapy for maintenance
- They increase salt and water excretion & reduce blood volume
 - Reduce preload & venous pressure
 - Improve cardiac performance & relieve edema

Diuretics for HF

- Aldosterone antagonist & other potassium sparing diuretic
↓ mortality by 30% when added to in patients receiving
ACE inhibitors and other standard therapy.
- Monitoring parameters: Potassium within 1 week of
treatment.

Vasodilators

- Nitroglycerin
- Nitroprusside
- After diuretics, intravenous vasodilators are the most useful medications for the management of acute HF.
- By stimulating guanylyl cyclase within smooth-muscle cells, nitroglycerin & nitroprusside dilating effects on arteries and veins.

Vasodilators

- results in a lowering of LV filling pressure, a reduction in mitral regurgitation, and improved forward cardiac output, without increasing heart rate or causing arrhythmias.
- The most common side effect of intravenous or oral nitrates is headache, which, if severe, can be treated with analgesics and often resolves during continued therapy.

Digoxin in HF

- Positive inotropic effect.
- Initiation: 0.125 mg-0.25 mg
 - Target serum concentration: 0.5-1.0 ng/mL
 - Monitoring parameters: digoxin toxicity

DIGITALIS

PHARMACOKINETICS:

	<i>DIGOXIN</i>	<i>DIGITOXIN</i>
❖ LIPID SOLUBILITY	MEDIUM	HIGH
❖ ORAL BIOAVAILABILITY	75%	>90%
❖ HALF-LIFE	40 HRS	168 HRS
❖ PLASMA PROTEIN BINDING	20-40 HRS	>90 HRS
❖ PERCENTAGE METABOLIZED	<20	>80

PHARMACOKINETICS:

*digitalis must be present in the body in certain "saturating" amount before any effect on congestive failure is noted .

This is achieved by giving a large initial dose in a process called "**digitalization**" -after initial dosages, digitalis is given in "maintenance" amounts sufficient to replace that which is excreted to avoid exceeding therapeutic range during digitalization:

- slow digitalization (over 1 week) is the safest technique
- plasma digoxin levels should be monitored

Mechanism of action

- ▶ Inhibit the monovalent cation transport enzyme coupled Na^+ , K^+ ATPase & increased intracellular Na^+ content $\Rightarrow$ increases intracellular Ca^{2+} through a Na^+ - Ca^{2+} exchange carrier mechanism.
- ▶ Increased myocardial uptake of Ca^{2+} augments Ca^{2+} release to the myofilaments during excitation $\Rightarrow$ invokes a positive inotropic response

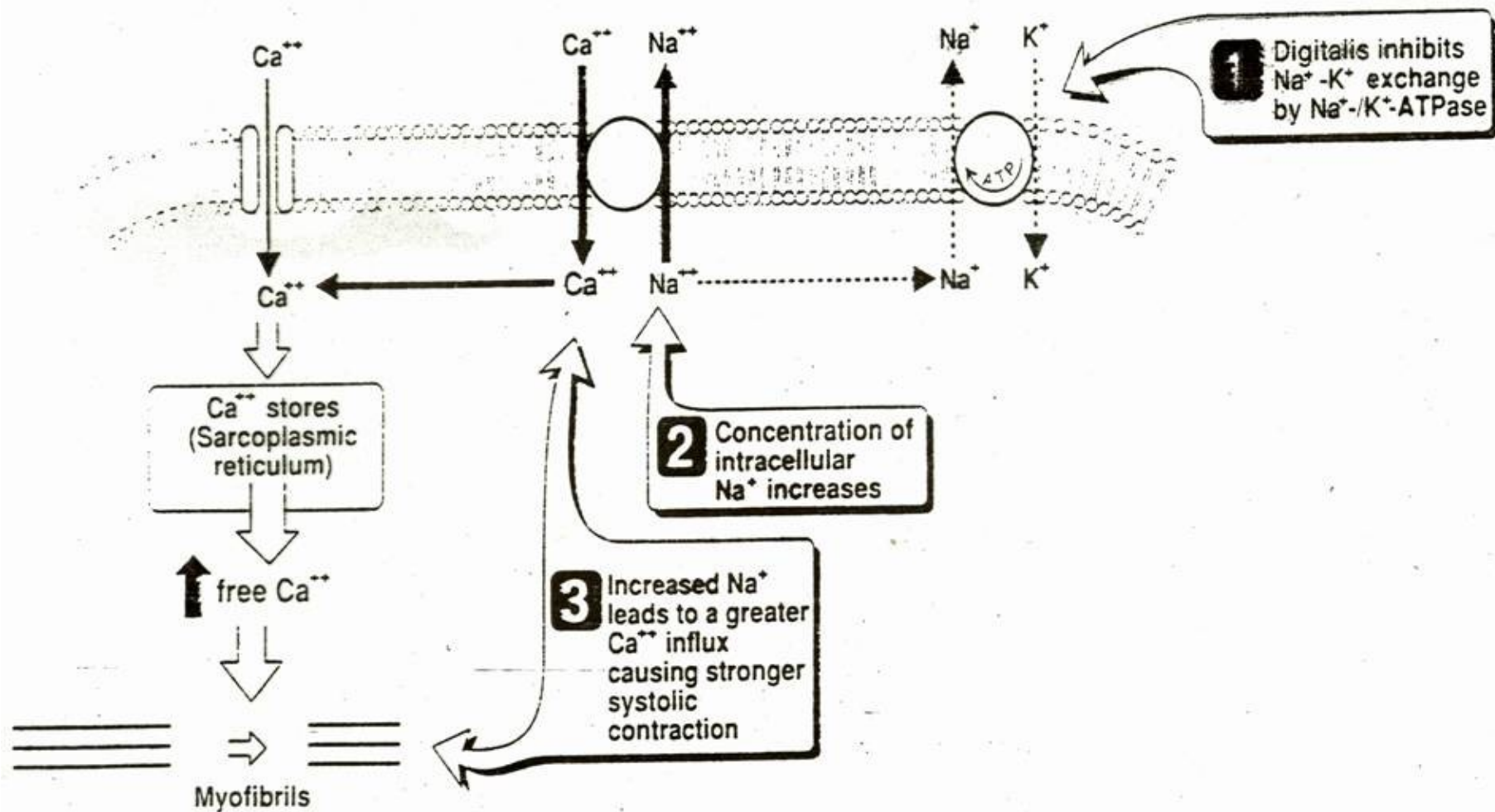


Figure 16.5
Mechanism of action of cardiac glycosides, or digitalis.

Effects of digitalis in heart failure

- ▶ Stimulates myocardial contractility
- ▶ Improves ventricular emptying
- ▶ Increase cardiac output
- ▶ Augments ejection fraction

DIGITALIS INTOXICATION:

- ▶ Serious & potentially fatal complication
- ▶ Anorexia, nausea & vomiting = earliest signs of digitalis intoxication
- ▶ Arrhythmias: Ventricular & atrial tachycardia w/ variable AV block
- ▶ Chronic digitalis intoxication = exacerbations of heart failure & weight loss

TREATMENT OF DIGITALIS INTOXICATION:

- Tachyarrhythmias: withdrawal of the drug, treatment with beta blocker or lidocaine
- Hypokalemia: potassium administration by the oral route
- **An antidote digoxin immune Fab (Digibind)**

Positive inotropic drugs

β -Adrenoceptor agonists

- They are used intravenously in CHF emergencies

Example of β -Adrenoceptor agonists :

Dobutamine

Exciting β_1 Adrenoceptor → positive inotropic action
→the volume of output↑

- Exciting β_2 Adrenoceptor → dilate the vascular → afterload↓

- have benefits within short time

Drugs Used for the Treatment of Angina (Anti-angina)

Objectives

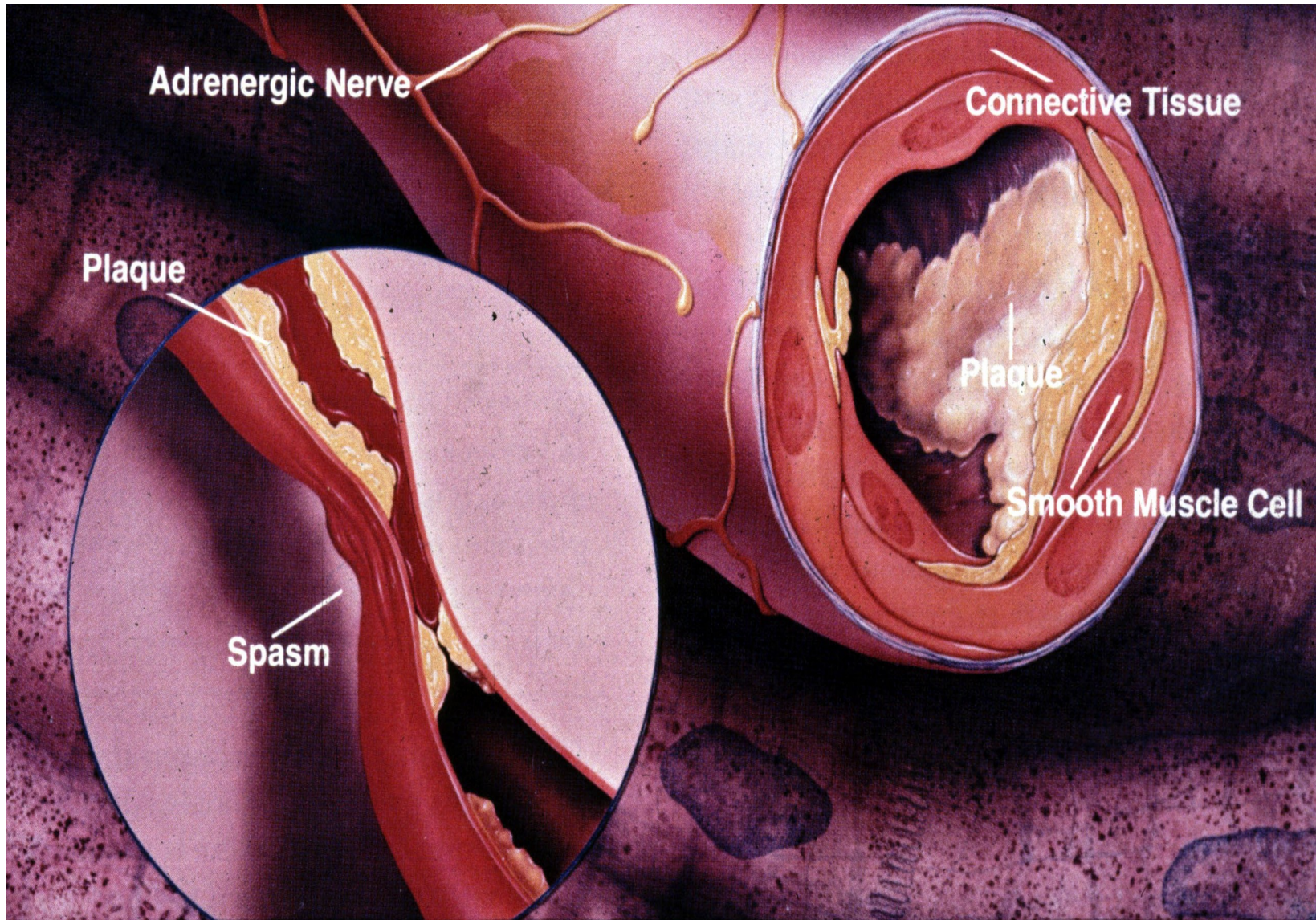
- Discuss about the pathophysiology of angina
- Describe the determinants of myocardial oxygen demand
- Discuss about the managements of angina

Definition

Angina pectoris – is chest pain due to ischemia of the heart muscle, generally due to obstruction or spasm of the coronary arteries.

- ✓ The main cause of Angina pectoris is Coronary Artery Disease, due to atherosclerosis of the arteries feeding the heart.

Ischemia may be defined as lack of oxygen and decreased blood flow in the myocardium.



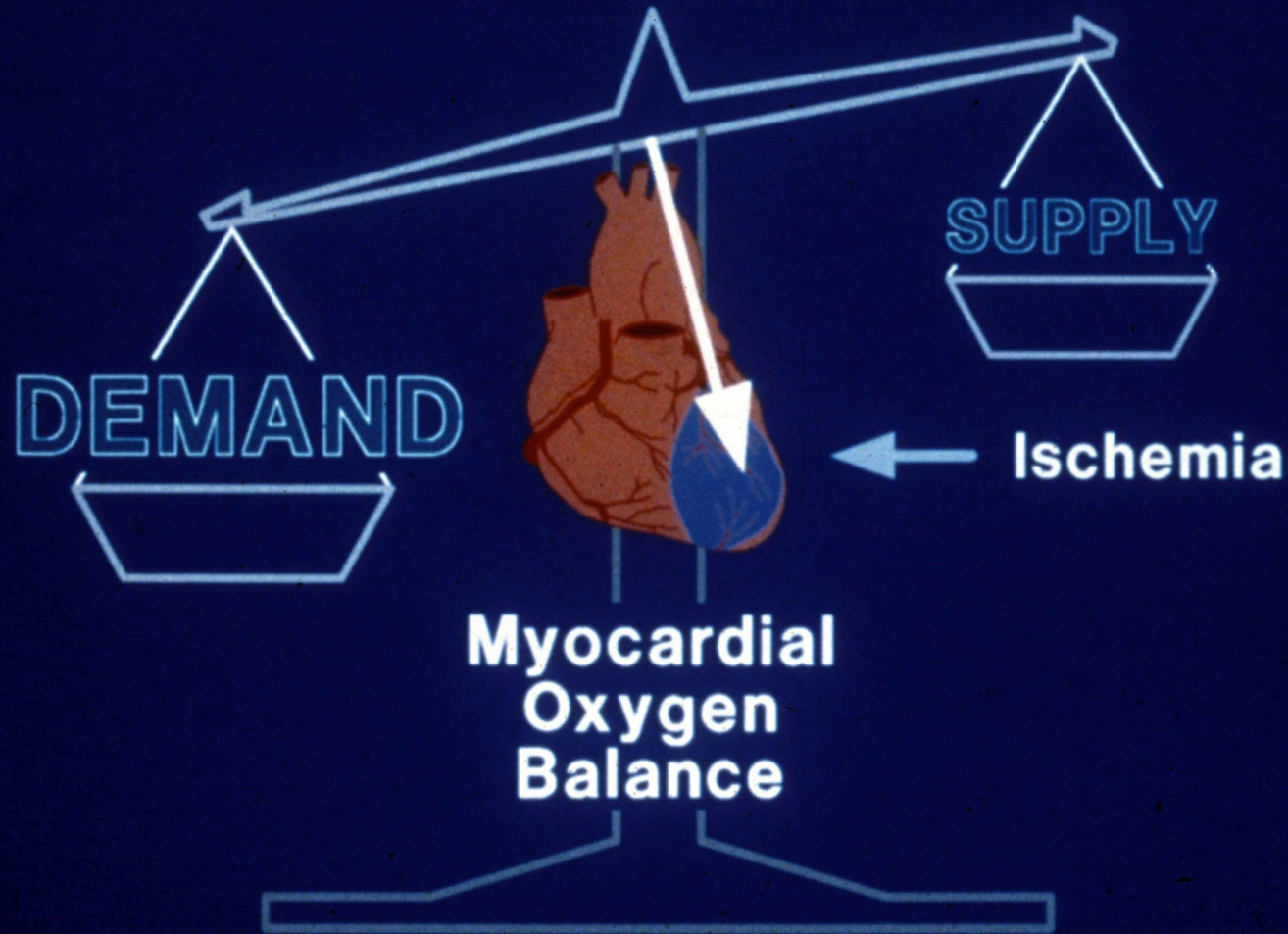
Pathophysiology of ischemic heart disease

- Angina pectoris is caused by transient episodes of myocardial ischemia due to an imbalance in myocardial oxygen supply and demand that may result from
 - increase in myocardial oxygen demand or,
 - a decrease in myocardial oxygen supply, or
 - sometimes from both.

Pathophysiology of ischemic heart disease

- All approved antianginal agents used in the treatment of angina such as nitrovasodilators, *beta adrenergic* receptor antagonists, Ca^{2+} channel antagonists, and antiplatelet agents improve the balance of myocardial oxygen supply and demand,
 1. by dilating the coronary vasculature or
 2. by reducing cardiac work

ANGINA PECTORIS IS A CONSEQUENCE OF MYOCARDIAL OXYGEN DEMAND EXCEEDING MYOCARDIAL OXYGEN SUPPLY



Determinants of myocardial oxygen demand

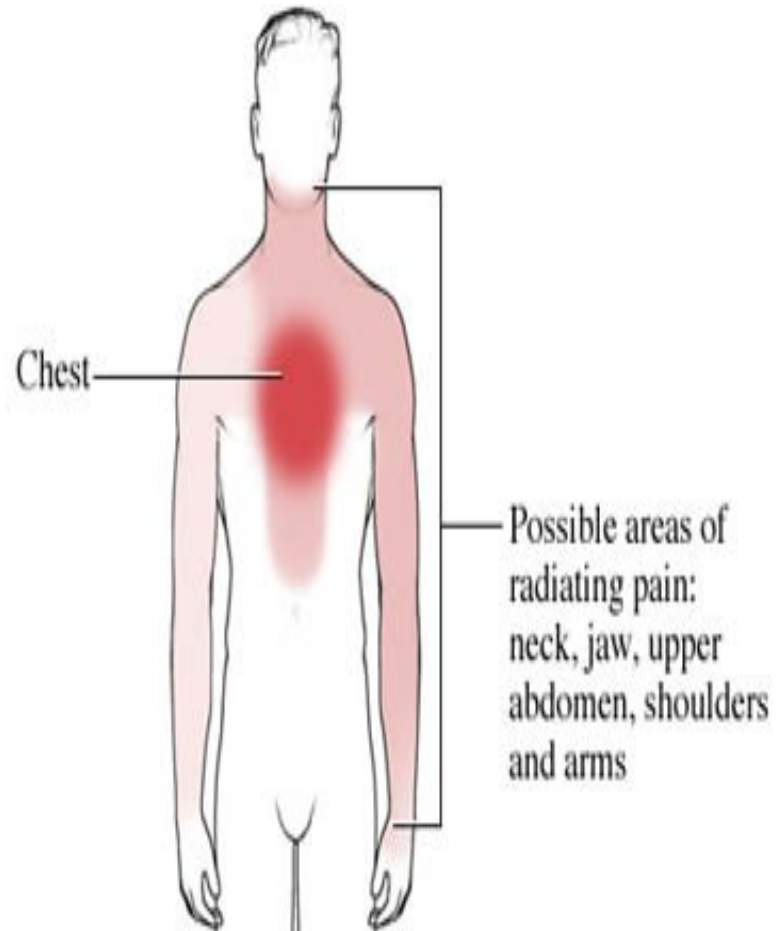
- Wall stress
- Intraventricular pressure
- Ventricular radius (volume)
- Wall thickness
- Heart rate
- Contractility

Major risk factors

- Age (≥ 55 years for men, ≥ 65 for women)
- Cigarette smoking
- Diabetes mellitus (DM)
- Dyslipidemia
- Family history of premature cardiovascular disease (men <55 years, female <65 years old)
- Hypertension (HTN)
- Kidney disease (microalbuminuria or $\text{GFR} < 60 \text{ mL/min}$)
- Obesity ($\text{BMI} \geq 30 \text{ kg/m}^2$)
- Physical inactivity

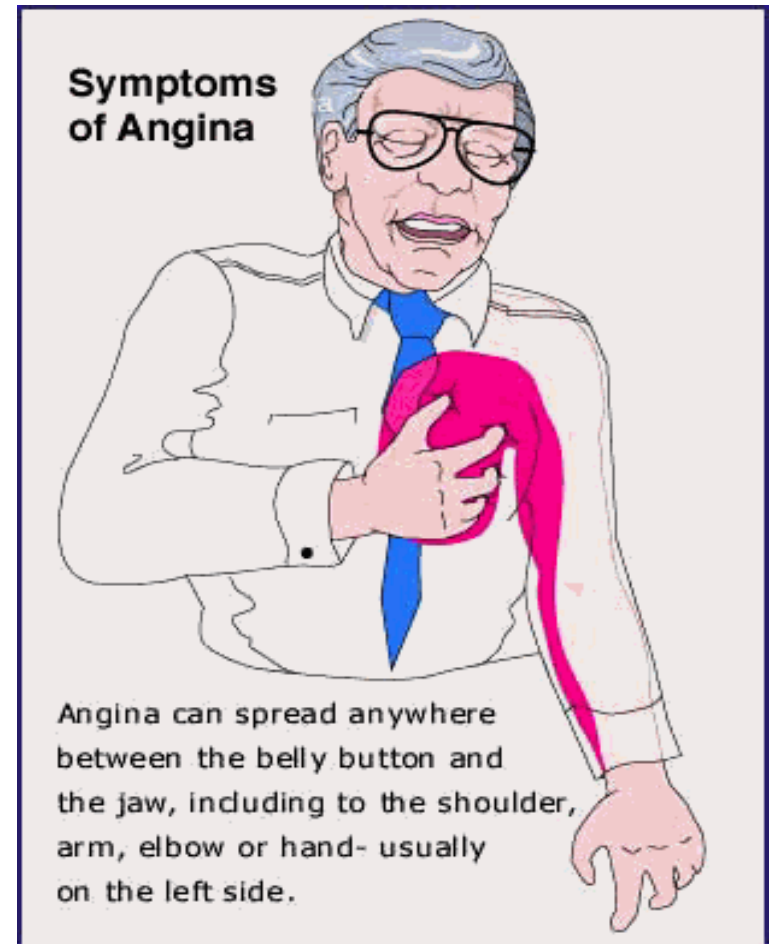
Types of Angina

- Angina occurs in three overlapping patterns:
 - Stable angina
 - Unstable angina
 - Prinzmetal (variant) angina



Stable Angina

- Attacks are predictably provoked by exercise, emotion, eating or coitus and subside when the increased energy demand is withdrawn.
- The underlying pathology is-severe arteriosclerotic affliction of larger coronary arteries .



Stable Angina

- Angina indicates that myocardial oxygen demand is exceeding supply.
- “Stable” indicates the reproducible nature of the angina; the same activity at the same intensity faithfully produces symptoms.
- Typically this type of angina is relieved by rest or acute use of nitroglycerin

Unstable Angina

- Unstable angina occurs when anginal symptoms occur with less cardiac demand.
- This is characterized by Pain that occurs with less exertion , cumulating pain at rest.
- Characterized by an acute or subacute worsening in a patient's anginal symptoms.



Unstable Angina

- Unstable angina with rapid increase in duration and severity of attacks is mostly due to rupture of an atheromatous plaque
- Atheromatous plaque attract platelet deposition and progressive occlusion of the coronary artery; occasionally with associated coronary vasospasm.

Prinzmetal (Variant) Angina

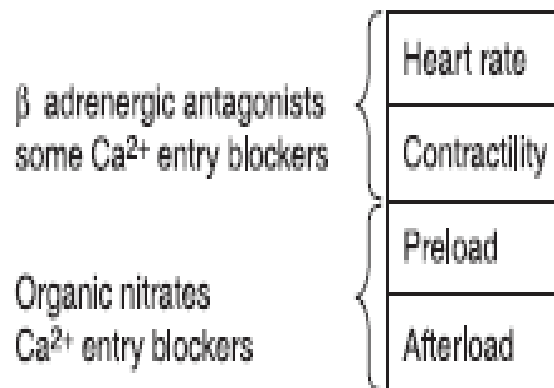
- It is induced by coronary artery vasospasm it generally responds promptly to vasodilators.
- PVA has been associated with other vasospastic disorders such as migraine headaches and Raynaud's phenomena.

Prinzmetal (Variant) Angina

- Endothelial dysfunction has been considered as primarily responsible for PVA.
- The Large Coronary Artery Spasm Usually associated with atheromatous disease.
- Attacks occur at rest or during sleep and are unpredictable.

Antianginal drugs

Agents decreasing O₂ Demand



Agents increasing O₂ Supply

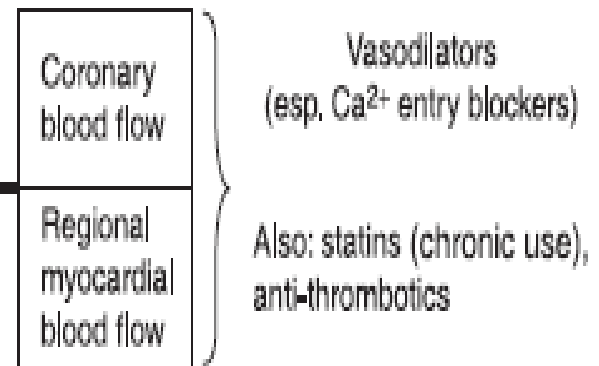


FIGURE 31-1 *Pharmacological modification of the major determinants of myocardial O₂ supply.* When myocardial O₂ requirements exceed O₂ supply, an ischemic episode results. This figure shows the primary hemodynamic sites of actions of pharmacological agents that can reduce O₂ demand (*left side*) or enhance O₂ supply (*right side*). Some classes of agents have multiple effects (*see text*). Stents, angioplasty, and coronary bypass surgery are mechanical interventions that increase O₂ supply. Both pharmacotherapy and mechanotherapy attempt to restore a dynamic balance between O₂ demand and O₂ supply.

Antianginal drugs

1. Nitrates

(Organic nitrates)

- Organic nitrates & nitrites are simple nitric & nitrous esters of glycerol.
- These agents cause a rapid decrease in myocardial oxygen demand leading to rapid resolution of symptoms.
- Nitrates are effective for all types of angina.

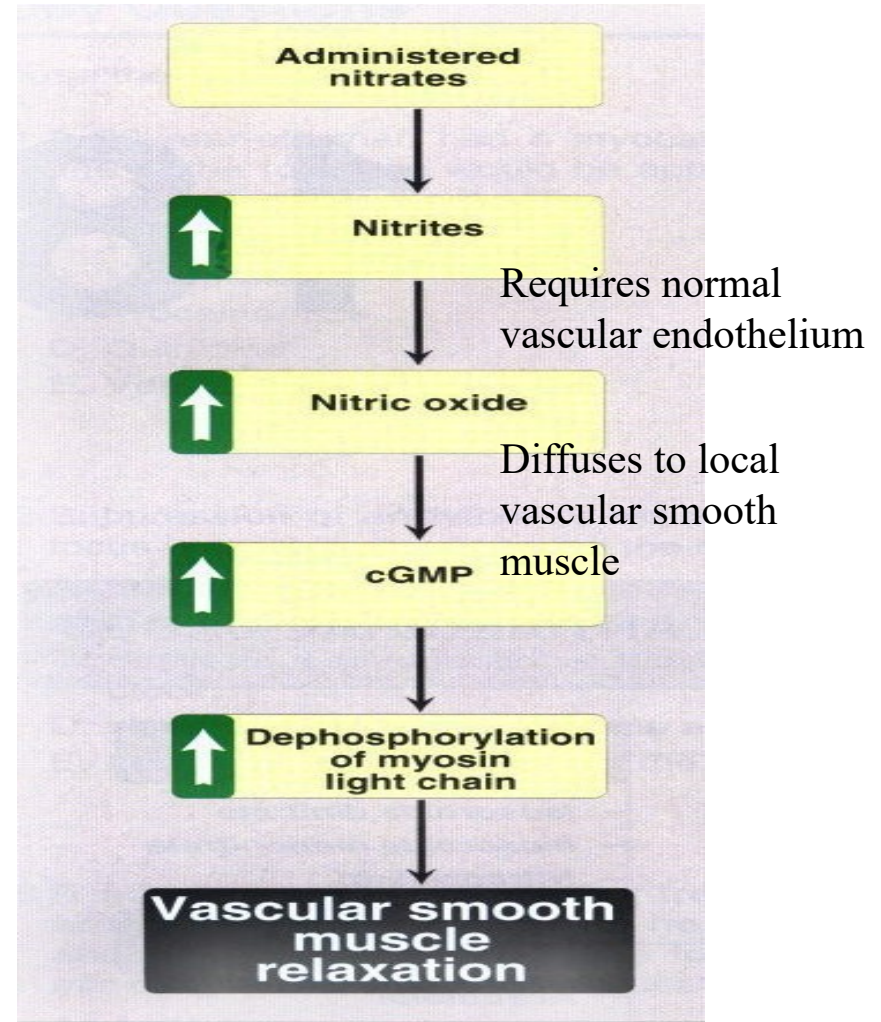


Figure 18.2

Effects of nitrates and nitrites on smooth muscle. cGMP = cyclic guanosine 3', 5'-monophosphate

Mechanism of Action

Nitrates decrease myocardial oxygen demand:

1. The primary effect is a reduction in venous tone which results in venous pooling decreasing venous return (decreased preload).
2. Arteriolar tone is less effectively reduced resulting in a decrease in PVR (decreased afterload) and decreased blood pressure.
3. #s 1 & 2 decrease myocardial wall stress reducing O_2 demand.
4. Dilation of coronary vessels or exerts a ~minor effect on increasing O_2 supply.

Adverse effects

1. The most common side effect of nitrates is headache.
2. Postural hypotension & syncope particularly with sublingual use.
3. Tachycardia induced by decreased PVR may itself induce anginal symptoms especially with unstable symptoms.

Adverse effects

Tolerance

- Sustained treatment with nitroglycerin in vivo is associated with reduced biotransformation of nitrate to NO by endothelial mitochondrial enzyme aldehyde dehydrogenase-2.
- Tolerance can be avoided by providing a “nitrate free” interval daily;

β -Blockers

- β -Blockers decrease O_2 demands of the myocardium by lowering the heart rate and contractility (decrease CO) particularly the increased demand associated with exercise.
- They also reduce PVR by direct vasodilation of both arterial & venous vessels reducing both pre- and after load.

β -Blockers

- Selective β_1 antagonists (atenolol, metoprolol and acebutolol) at higher dose cause bronchospasm.
- β_1 antagonists reduce the frequency and severity of anginal episodes particularly when used in combination with nitrates.
- β -Blockers in combination with nitrates can be quite effective.

Ca²⁺ Channel Blockers

- Ca²⁺ channel blockers protect tissue by inhibiting the entrance of Ca²⁺ into cardiac and smooth muscle cells of the coronary and systemic arterial beds.
- All Ca²⁺ channel blockers produce some vasodilation (↓ PVR) and (-) inotropes.
- May exacerbate CHF .
- They are useful in Prinzmetal angina in conjunction with nitrates.

Ca⁺² Channel Blockers

Agents

1. Nifedipine: This Ca⁺² channel blocker works mainly on the arteriolar vasculature decreasing after load
 - It has minimal effect on conduction or HR.
 - It is metabolized in the liver and excreted in both the urine & the feces.
 - It causes flushing, headache, hypotension and peripheral edema.
 - Constipation.
 - A reflex tachycardia associated with the vasodilation.

Dose: 10–20 mg TID PO; may increase to 120 mg/d.

Ca⁺² Channel Blockers

2. Verapamil: The agent has its main effect on cardiac conduction decreasing HR and thereby O₂ demand.
- It also has much more (-) inotropic effect than other Ca⁺² channel blockers.
 - It is a weak vasodilator.
 - It is metabolized in the liver.
- Dose: 40-760 mg TID oral, 5 mg by slow iv injection

Partial fatty acid oxidation inhibitors

Ranolazine

- Is newer antianginal drug approved by US-FDA in 2006 for treatment of chronic angina pectoris in patients who fail to respond to standard antranginal therapy.
- Acts by partially inhibiting fatty acid oxidation in the myocardium, thus shifting metabolism to glucose which requires less oxygen to metabolize
- – QT prolongation and testicular toxicity are among the possible toxicities so far.

Dose:0.5-7.0 g BID

Combination therapy

When monotherapy is unable to provide adequate relief in tolerated doses, concurrent use of 2 or 3 drugs may be tried.

I. B blocker + long-acting nitrate combination is rational in classical angina because:

(a) Tachycardia due to nitrate is blocked by B blocker.

(b) The tendency of B blocker to reduce total coronary flow is opposed by nitrate.

II. Nitrates with CCB

- Nitrates primarily decrease preload, while CCBs have a greater effect on afterload. Their concurrent use may decrease cardiac work to an extent not possible with either drug alone.
- This combination may be especially valuable in severe vasospastic angina.

Combination therapy

III. B blocker + Nitrate + CCB

- Nitrates primarily decrease preload.
- CCBs mainly reduce afterload + increase coronary flow.
- B blockers decrease cardiac work primarily by direct action on heart.

Mechanical interventional(mechanotherapy)

1. percutaneous coronary intervention

- Percutaneous coronary intervention is a non-surgical method used to open narrowed arteries that supply heart muscle with blood (coronary arteries).
- Percutaneous means "through unbroken skin."
- Percutaneous coronary intervention is performed by inserting a catheter through the skin in the groin or arm into an artery.

Mechanical interventional(mechanotherapy)

- At the leading tip of this catheter, several different devices such as a balloon ,cutting device (arthrectomy device) can be deployed.
- The catheter and its devices are threaded through the inside of the artery back into an area of coronary artery narrowing or blockage.

Mechanical interventional(mechanotherapy)

- Percutaneous coronary intervention can be used to relieve or reduce
 - angina,
 - alleviate congestive heart failure, and
 - allows some patients to avoid surgical treatment (coronary artery bypass graft or CABG) that involves extensive surgery and often long rehabilitation time.

Mechanical interventional(mechanotherapy)

2. Coronary Artery Bypass Graft surgery

- Coronary artery bypass grafting (CABG) is a type of surgery that improves blood flow to the heart.
- Surgeons use CABG to treat people who have severe coronary heart disease (CHD) mainly due to atherosclerosis.
- In people with atherosclerosis, Over time, plaque can harden or rupture (break open).

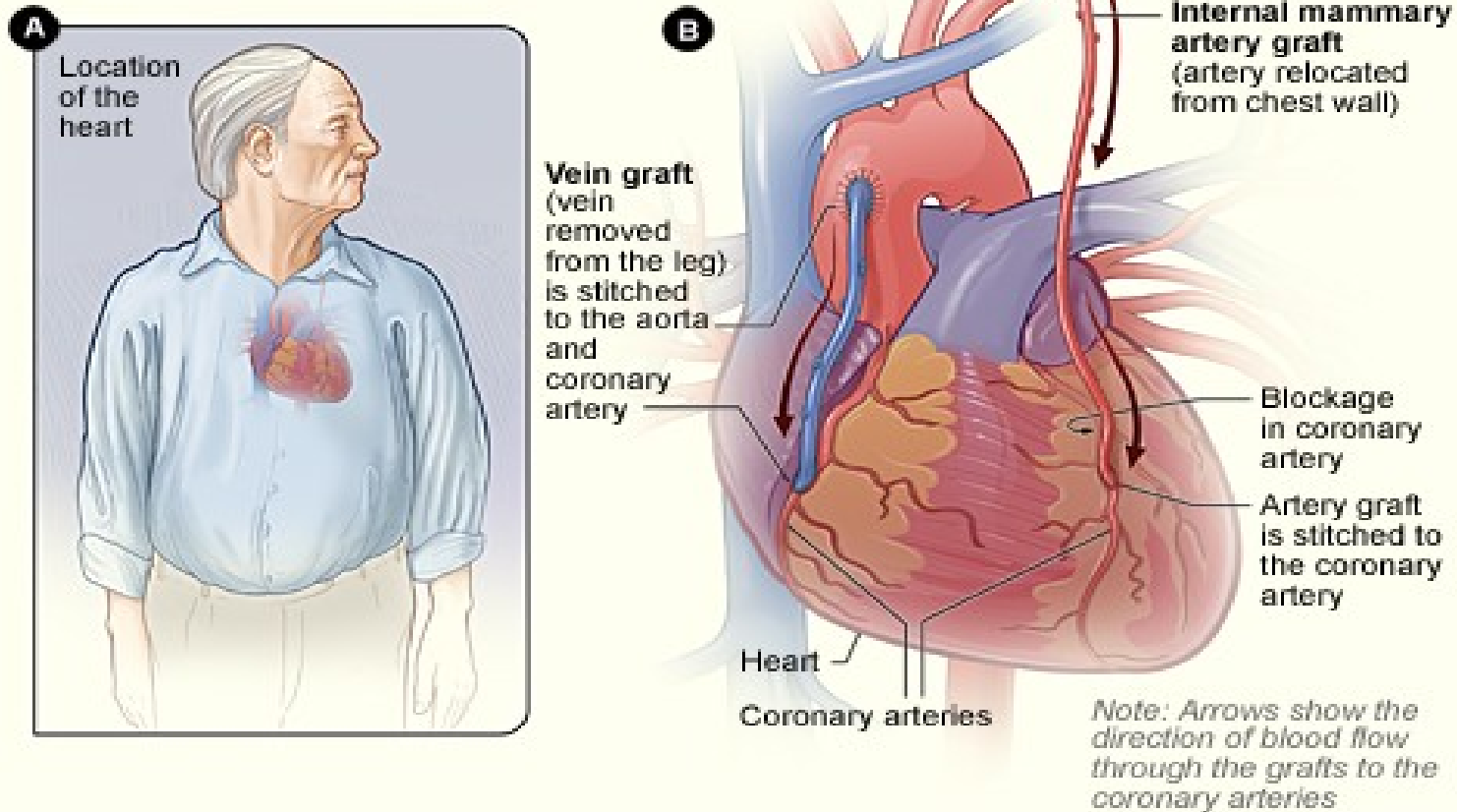
Mechanical interventional(mechanotherapy)

- Hardened plaque narrows the coronary arteries and reduces the flow of oxygen-rich blood to the heart.
- This can cause chest pain or discomfort called angina.
- If the plaque ruptures, a blood clot can form on its surface.
- A large blood clot can mostly or completely block blood flow through a coronary artery.
- This is the most common cause of a heart attack.

Mechanical interventional(mechanotherapy)

- CABG is one treatment for CHD.
- During CABG, a healthy artery or vein from the body is connected, or grafted, to the blocked coronary artery.
- The grafted artery or vein bypasses (that is, goes around) the blocked portion of the coronary artery. This creates a new path for oxygen-rich blood to flow to the heart muscle.
- Surgeons can bypass multiple coronary arteries during one surgery.

Coronary Artery Bypass Grafting



Anti-hyperlipidemics and Anti-arrhythmic drugs

Objectives

- Describe the pathophysiology of atherosclerosis
- Discuss management of hyperlipidemia
- Explain mechanisms of cardiac Arrhythmias
- Describe electrophysiology of normal contraction of heart
- Discuss about anti-arrhythmics

Anti-hyperlipidemics

Introduction

Hyperlipidemia

- Dyslipidemia is defined as elevated total cholesterol, LDL cholesterol, or triglycerides; a low HDL cholesterol; or a combination of these abnormalities.
- Triglycerides and cholesterol are essential constituents of the organism
- Triglycerides represent a form of energy store and cholesterol is a basic building block of biological membranes
- Abnormalities of plasma lipids can result in a predisposition to coronary, cerebrovascular, and peripheral vascular disease.
- The risk of CHD is directly proportional to blood cholesterol levels and a lowering of cholesterol, specifically LDL cholesterol, decreases the incidence of heart attacks.

Introduction

The principal lipids in the blood are:

- 1- Triglycerides (TG) .
- 2- Phospholipids.
- 3- Cholesterol .
- 4- Cholesterol esters .
- 5- Non-esterified fatty acids

Lipoprotein: Macromolecules that transport lipids in the plasma(a group of soluble proteins that combine with and transport fat or other lipids in the blood plasma.

They may be classified according to their descending densities into:

- 1-High density lipoprotein (HDL).
- 2-Low density lipoprotein (LDL).
- 3-Intermediate density lipoprotein (IDL).
- 4-Very low-density lipoprotein (VLDL).
- 5-Chylomicrons .

Introduction

- Chylomicrons transport fats from the intestinal mucosa to the liver
- In the liver, the chylomicrons release triglycerides and some cholesterol, and become low-density lipoproteins (LDL).
- LDL then carries fat and cholesterol to the body's cells.
- Liver requires cholesterol for synthesizing VLDL particles and bile acid
- High-density lipoproteins (HDL) carry fat and cholesterol back to the liver for excretion.
- When oxidized LDL cholesterol gets high, atheroma formation in the walls of arteries occurs, which causes atherosclerosis.
- HDL cholesterol is able to go and remove cholesterol from the atheroma.
- Atherogenic cholesterol → LDL, VLDL, IDL

Causes of Hyperlipidemia

- Diet
- Hypothyroidism
- Acute hepatitis
- Anorexia nervosa
- Obstructive liver disease
- Obesity
- Diabetes mellitus
- Pregnancy

Dietary sources of Cholesterol

Type of Fat	Main Source	Effect on Cholesterol levels
Monounsaturated	Olives, olive oil, canola oil, peanut oil, cashews, almonds, peanuts and most other nuts; avocados	Lowers LDL, Raises HDL
Polyunsaturated	Corn, soybean, safflower and cottonseed oil; fish	Lowers LDL, Raises HDL
Saturated	Whole milk, butter, cheese, ice cream; red meat; chocolate; coconuts, coconut milk, coconut oil , egg yolks, chicken skin	Raises both LDL and HDL
Trans	Most margarines; vegetable shortening; partially hydrogenated vegetable oil; deep- fried chips; many fast foods; most commercial baked goods	Raises LDL

Pathophysiology of atherosclerosis

- Lipids are transported in the bloodstream as **complexes** of lipid and proteins known as lipoproteins.
- Atherosclerosis can result from **injury to endothelium** accompanied with or mediated by **oxidation; infection ; or a combination of those**.
- **Oxidized** LDL provokes an **inflammatory response** mediated by a number of chemoattractants and cytokines.
- **Repeated** injury and repair within an atherosclerotic plaque eventually lead **plaque rupture** and subsequent coronary thrombosis may occur.

- Diagnosis is typically based on medical history, physical examination and blood test done after overnight fasting

LIPID PROFILE			
	DESIRABLE	BORDERLINE	HIGH RISK
Cholesterol	<200 mg/dl	200-239 mg/dl	240 mg/dl
Triglycerides	<150 mg/dl	150-199 mg/dl	200-499 mg/dl
HDL cholesterol	60 mg/dl	35-45 mg/dl	<35 mg/dl
LDL cholesterol	60-130 mg/dl	130-159 mg/dl	160-189 mg/dl
Cholesterol/ HDL ratio	4.0	5.0	6.0

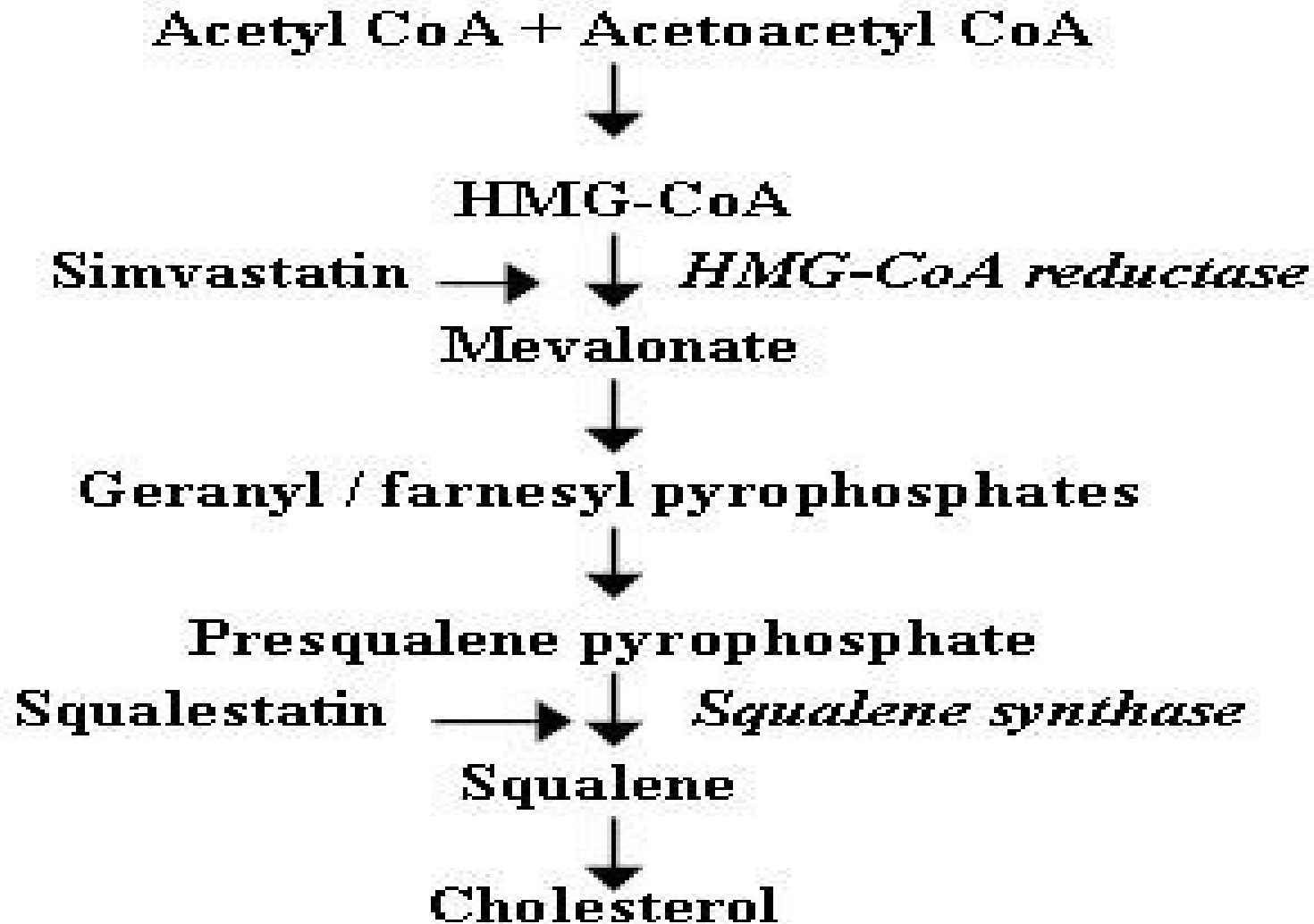
1. Therapeutic lifestyle changes (TLC)

- Initiating and maintaining TLC is an important part of all lipid-lowering regimens and can be highly cost-effective when adhered to long term.
- Components of TLC that have been shown to be effective in lowering LDL-C include a **healthy diet(a cholesterol-lowering diet)**, regular physical activity, smoking cessation, control hypertension and weight loss.

2. Drug therapy

- Circulating cholesterol is derived from two sources: ***de novo synthesis in*** peripheral tissues and ***intestinal absorption***.
- ***Each source contributes*** approximately half of the total cholesterol found in the blood.

Cholesterolatherosclerosis



Anti-hyperlipidaemic drugs

[I] Agents targeting endogenous cholesterol synthesis

1) HYDROXYMETHYLGLUTARYL CO-ENZYME A (H M G - COA)REDUCTASE INHIBITORS (STATINS):

Preparation:

- Lovastatin , Simvastatin, Pravastatin, Atorvastatin, Fluvastatin, Cerivastatin
- The liver is their target organ, and decreased hepatic cholesterol synthesis ultimately leads to increased removal of LDL particles from the circulation.

Mechanism of action

- These are potent reversible competitive inhibitors of 3-hydroxy 3-methyl glutaryl coenzyme A reductase, the rate-controlling enzyme in cholesterol biosynthesis.

Pharmacological action

- ✓ **Effect on LDL-C:** *Statins decrease LDL-C by two mechanisms:*
 1. Up-regulation of LDL-R with increase of clearance of LDL-C and decrease LDL-C.
 2. Decrease of very low density lipoprotein (VLDL) production [C is a required component of VLDL] which is a precursor of LDL-C
- ✓ **Effect on HDL-C:** Statins induce modest increase in HDL-C

Adverse effects

- 1- Hepatotoxicity.
- 2- Myopathy increased incidence with drug combination.
 - a- With other lipid lowering drugs(inhibit statin uptake in to the liver)
 - i) Fibrates. ii) Niacin.
 - b- With other drugs that are metabolized by 3A4(CYP3A4 inhibitors)
isoform of cytochrome P450 e.g.: erythromycin, cyclosporine, verapamil, ketoconazole.

Pravastatin is the drug of choice to be given with verapamil, cyclosporine ketoconazole because it is not metabolized by cytochrome 3A4 system.

Therapeutic uses

- Hypercholesterolemia

Contraindication:

- 1- Pregnancy.
- 2- Active liver diseases.

Drug interaction :

- They potentiate the action of oral anticoagulant

2 Fibric acid derivative (fibrates)

Preparations:

- Gemfibrozil, Clofibrate

Mechanism of action:

- 1- Increased LPL(key enzyme in the degradation of VLDL) activity, which increase clearance of TG from LP in the plasma.
- 2- Increased FFA uptake by the liver.
- 4- Increased LDL-C uptake by the liver.
- 5- Raises HDL cholesterol levels

Pharmacological action of fibrates

✓ **Effect on VLDL:** Decreased VLDL due to:

- 1) Decreased synthesis in liver.
- 2) Increased clearance.

✓ **Effect on LDL:** Decreased LDL due to:

- 1) Decreased LDL precursor (VLDL)
- 2) Increased LDL uptake.

✓ **Effect on HDL:** Increased HDL.

Therapeutic uses:

- Fibrates are the drugs of choice in treatment of hypertriglyceridaemias.

Drug interaction:

- Clofibrate increases the sensitivity to coumarins anticoagulant by displacing them from their protein binding sites.
- Clofibrate also causes potentiation of the hypoglycaemic action of sulphonylurea drugs.

Contraindications:

- 1- Patients with impaired renal or hepatic functions.
- 2- Pregnant or nursing women.

3 Nicotinic acid(Niacin)

Mechanism of action:

1. In adipose tissue: it decreases free fatty acids mobilization from adipocytes.
2. In liver: it decreases VLDL production(believed to be to reduced flow of fatty acids from adipose tissue to liver (decreased TG synthesis).
3. In plasma: it increases LPL activity that increases clearance of VLDL & chylomicron.

Pharmacological action:

- **Effect on VLDL:** Decreased VLDL by:
 - 1) decreased synthesis in liver;
 - 2) increased clearance in plasma.

Pharmacological action

Effect on LDL: Decreased LDL due to reduction in its precursor (VLDL).

Effect on HDL: Induces modest increase in HDL-C

Therapeutic uses:

1. Hypercholesterolemia in which both LDL&VLDL are increased.
2. Patient with hypertriglyceridemia & low HDL-C.
3. Hyperchylomicronemia.
4. Combined hyperlipidemia.

Adverse effects of niacin

- Pruritus, flushing (which is prostaglandin mediated avoided by low dose aspirin 325 mg ½ h before niacin).
- Dypepsia: nausea, vomiting (decreased if taken after meal).
- Reactivation of peptic ulcer.
- Hepatotoxicity.
- Increased uric acid level.

Contraindications

- Gout
- Peptic ulcer
- Hepatotoxicity

Drug interaction

- Postural hypotension may occur in patients on antihypertensives when they take nicotinic acid

II-Agents Targeting Exogenous Cholesterol

Cholesterol Uptake Inhibitors:

1- Ezetimibe

Mechanism of action:

- A new class of drugs that specifically blocks absorption of intestinal cholesterol.
- Approximately 50% of the total cholesterol found in the gut after a meal is absorbed by the intestinal enterocyte and is passed into the circulation in the form of a triglyceride-rich chylomicron.
- The unabsorbed cholesterol is eliminated with the feces. Ezetimibe inhibits as much as 54% of all intestinal cholesterol absorption without affecting uptake lipid-soluble vitamins.

Pharmacological action

Effect on LDL-C: It decreases LDL due to decrease in intestinal absorption of cholesterol.

Therapeutic uses:

- Ezetimibe can be used as mono-therapy in patients at low risk of CHD who require a modest reduction in their LDL level or in patients who cannot tolerate statin therapy.
- Co-administration of ezetimibe and a statin reduces both intestinal cholesterol absorption and hepatic cholesterol synthesis, resulting in a significant reduction of the plasma LDL-C level.

Adverse effect

- No specific adverse effect, except reversible hepatic dysfunction and rarely myositis

2- Bile acid binding resins(bile acid sequestrants)

Mechanism of action:

- These resins, given orally, are not absorbed. They bind bile acids in the intestine and they increase their fecal excretion. Since bile acids inhibit the enzyme that catalysis the rate limiting step in the conversion of cholesterol to bile acids, their removal results in increased breakdown of hepatic cholesterol.
- In addition, since bile acids are required for intestinal absorption of cholesterol, these resins decrease cholesterol absorption from the G.I.T. However, a compensatory increase occurs in the rate of biosynthesis of cholesterol which is insufficient to compensate for the increased catabolism and thus the plasma and tissue cholesterol levels decrease.

Chemistry

- They are highly positively charged anions that bind negatively charged bile acids.

Pharmacological action:

- **Effect on LDL-C:** It decreases C content of hepatocytes leading to upregulation of LDL-R with increased LDL-C clearance from blood and decreased LDL-C level.

Effect on HDL-C: Increased HDL-C

1) Constipation. 2) G.I.T complaints: heart burn, flatulence, dyspepsia.

Drugs interaction:

- May bind other drugs and interfere with their absorption:
 - Thiazides · Furosemide · Digoxine
 - Propranolol · L-thyroxin · Some statins
 - Coumarine anticoagulants
- These drugs are taken one hour before or four hours after resins.

Contraindications

- 1- Biliary obstruction
- 2- Chronic constipation

3- Beta - sitosterol (cytelin)

- It is a plant sterol chemically similar to cholesterol. Sitosterol interferes with cholesterol absorption from the intestine by competition.
- Consequently, it decreases absorption of both exogenous cholesterol from the diet and endogenous cholesterol from the bile.
- It is usually used in combination with other drugs.

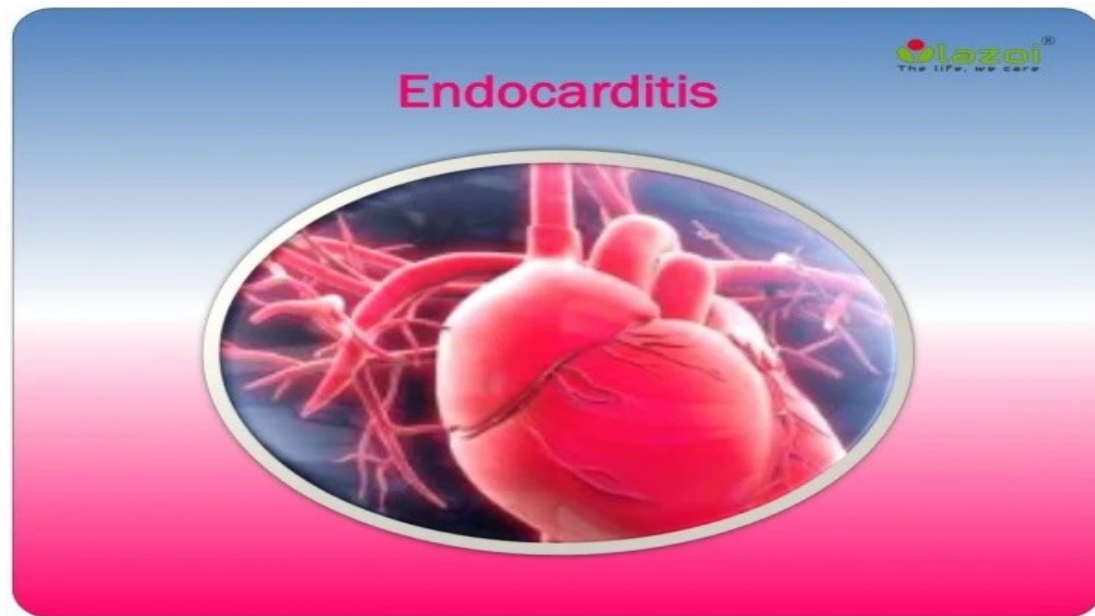
Dosage: 5 g orally before each meal

Combination therapy

Indication:

- Combined increase in LDL & VLDL.
- High LDL or VLDL not normalized with a single drug.
- Severe hypertriglyceridemia or hypercholesterolemia.
- To take lower doses of each drug.

Pharmacology of endocarditis



Introduction

- Endocarditis is an infection of the inner lining of the heart and its valves.
- The infection is caused by bacteria or fungi that enter the bloodstream through the mucous membranes of the mouth, open wounds, needles. This condition can also happen from medical procedures which involve cutting through infected skin.
- Endocarditis can lead to other life-threatening conditions. Stroke or reduced blood flow to other organs can occur when clumps of bacteria break off from the heart and block the blood vessels.
- The bacteria that cause endocarditis can also spread to other organs, causing additional tissue damage. Uncontrolled endocarditis can permanently damage the heart valves, eventually leading to heart failure.

Classification

Two categories: acute or sub acute [older classification]

- Acute disease
 - Often : virulent organisms (S. aureusaureus).
 - More aggressive/Toxic presentation
 - Progressive valve destruction & metastatic infection developing in days to weeks
 - High fevers, leukocytosis, and systemic toxicity
 - Mortality: few days to weeks.
- Sub-acute disease
 - Less virulent organisms (viridans group streptococci)
 - Produce a slower and more subtle presentation.
 - Weakness, fatigue, low-grade fever, night sweats, weight loss
 - Death occurring in several months.
 - Metastasis less common

High risk factors

- Congenital or structural cardiac defects,
- Valvular disease, long-term
- hemodialysis,
- diabetes mellitus; poor oral hygiene,
- Major dental treatment, previous endocarditis,
- Hypertrophic cardiomyopathy,
- Mitral valve prolapse with regurgitation
- Any structural cardiac defect → turbulence of blood flow
→ → predisposes to the development of IE

Risk factors & preventions

- Patients who are having high risk of endocarditis should practice good dental hygiene, including regular brushing, flossing and visits to the dentist.
- The bacteria that cause endocarditis often enter the bloodstream through tissue damage in the mouth caused by poor dental hygiene (gingivitis).
- Patients who are having high risk of endocarditis should also avoid cosmetic procedures, such as piercing or tattooing, which can push bacteria from the skin into the bloodstream.
- Patients at high risk should also see a doctor if they have normal wounds that do not heal quickly.
- Finally, patients who are having high risk of endocarditis may be prescribed preventive antibiotics before some medical procedures, including dental work that involves cutting into the gums or teeth, or certain types of surgery.

Classification

- Can also classified as: [newer classification]
 - Native Valve IE
 - Prosthetic Valve IE
- Additional Consideration
- Intravenous drug abuse (IVDA) IE
- Nosocomial IE

Common etiologic agent

- Any type of microorganism
- In majority of cases: streptococci, staphylococci, & enterococci.
- Less common: Gm(-), fungal, and atypical organisms[HACEK*]

Diagnosis of Endocarditis

Tests commonly used to diagnose endocarditis include:

- Blood Tests Echocardiogram
- Electrocardiogram
- Chest X-rays, Computed Tomography and Magnetic Resonance Imaging (MRI)

Symptoms of Endocarditis

- The most common symptoms of endocarditis mimic the flu or other viral infections, including:
 - Fever
 - Chills
 - Sweating, night sweats
 - Lack of energy, weakness
 - Paleness
 - Aching joints and muscles

Treatment of Endocarditis

➤ Therapeutic Considerations

- Penetrate into the vegetation,
- Achieve adequate drug concentrations,
- Achieve adequate kill rates.
- Parenteral at high doses
- Bactericidal and synergistic activity
- Extended treatment course of 4 to 6 weeks

➤ Goal:

- Eradicate the infection and minimize/prevent any complications

Empirical therapy

- Evaluate for risk factors [hint for most likely causative organism].
- If no risk factors: cover gram-positive organisms primarily.
- Streptococci: penicillin plus gentamicin
- Staphylococci or enterococci: vancomycin plus gentamicin

Directed therapy

- DOC: penicillin G
- Ceftriaxone: alternative
- **Length of therapy:** 4 weeks
- If the shorter length of therapy is chosen, add gentamicin (ie, 2 weeks).
- **Vancomycin:** allergic or intolerant to β -lactams
- **Patients with PVE:** 6 weeks with penicillin G or ceftriaxone with or without gentamicin during the initial 2 wks of therapy.
- **Less susceptible:** penicillin G or ceftriaxone + gentamicin for the entire 6 weeks

Staphylococci IE

- First determine
 - a) Whether the isolate is methicillin susceptible versus methicillin resistant and
 - b) Patient has a native versus prosthetic valve.
- **Native valve [methicillin-susceptible]:** penicillinase-resistant penicillin (eg, nafcillin or oxacillin) or cefazolin with or without gentamicin,
- **Methicillin resistant:** add vancomycin
 - Add (1st days of 3-5 days) of aminoglycosides

Staphylococci IE

- Vancomycin + gentamicin for first 2 weeks/ rifampin for the entire length of treatment.
- Alternatives: Daptomycin, Ceftaroline, linezolid, quinupristin/dalfopristin, or tigecycline alone or in combination.
- Staphylococcal PVE: treat for minimum of 6 weeks

Enterococci IE

- High-dose penicillin G, ampicillin, or vancomycin + gentamicin
- Treatment length is usually 4 to 6 weeks, with the aminoglycoside used over the entire course.
- Vancomycin: Alternative
- VRE
 - Daptomycin, linezolid or quinupristin/dalfopristin for a minimum of 8 weeks.

Imipenem/cilastatin + ampicillin or **ceftriaxone** + ampicillin for the treatment of *E. faecalis*, [8 weeks of therapy

Directed therapy

- Gm(-) Organisms
 - **Pseudomonas spp.**
 - Antipseudomonal β -lactam (eg, piperacillin/tazobactam, cefepime, imipenem/cilastatin)+ high-dose aminoglycoside (tobramycin 8 mg/kg/day).
 - Length of treatment: minimum of 6 weeks.

HACEK Group

- Are difficult to isolate, often taking weeks for identification.
- Ceftriaxone (or 3rd /4th generation cephalosporin): DOC
- Ampicillin-sulbactam/ciprofloxacin: alternative
- Length of treatment: 4 weeks

Directed therapy

➤ Fungi

- Requires early valve replacement[often considered]
- Treatment of fungal IE is exceptionally difficult.
- IV Amphotericin B(0.6–1 mg/kg/day): DOC.IV
- with or without 5-flucytosine (25 mg/kg orally 4x/daily)
- Duration Rx: minimum of 8 weeks of treatment.
- Oral azoles (eg, fluconazole): as long-term suppressive therapy to prevent relapse

Directed therapy...

- **Surgery:** in
- Unresolved infection, ineffective antimicrobial therapy,
- More than one episode of serious emboli,
- Refractory congestive heart failure,
- Significant valvular dysfunction,
- Mycotic aneurysm requiring resection,
- Local complications (perivalvular or myocardial abscesses), or
- Prosthetic-valve infection associated with a pathogen demonstrating higher antimicrobial resistance

Prophylaxis Issues

- To prevent the development of IE,...appropriate for these at-risk patients:
 - Presence of prosthetic valves or material,
 - Prior IE, congenital cardiac disease and cardiac transplant
 - Dental procedures involving manipulation of gingival tissue or periapical region of teeth or perforation of the oral mucosa.
 - Prophylaxis for GI or genitourinary surgeries primarily targets enterococci

Pharmacology of pericarditis

Introduction

- Pericarditis is inflammation of the pericardium (the fibrous sac surrounding the heart)
- Pericarditis is inflammation of the pericardium, a sac-like structure with two thin layers of tissue that surround the heart to hold it in place and help it work.

Etiology

1. Idiopathic

2. Infections

- A. Viral - Coxsackievirus, echovirus, adenovirus, EBV, CMV, influenza, varicella, rubella, HIV, hepatitis B, mumps, parvovirus B19,
- B. Bacterial - Staphylococcus, Streptococcus, pneumococcus, Haemophilus, Neisseria (gonorrhoeae or meningitidis), Chlamydia (psittaci or trachomatis), Legionella, tuberculosis, Salmonella, Lyme disease
- Mycoplasma
- D. Fungal - Histoplasmosis, aspergillosis, blastomycosis, coccidioidomycosis, actinomycosis, nocardia, candida
- E. Parasitic - Echinococcus, amebiasis, toxoplasmosis
- F. Infective endocarditis with valve ring abscess

Etiology.....

3. Neoplasm

- A. Metastatic – Lung or breast cancer, Hodgkin's disease, leukemia, melanoma
- B. Primary – Rhabdomyosarcoma, teratoma, fibroma, lipoma, leiomyoma, angioma
- C. Paraneoplasm

4. Cardiac

- A. Early infarction pericarditis
- B. Late postcardiac injury syndrome (Dressler's syndrome)
- C. Myocarditis
- D. Dissecting aortic aneurysm

5. Autoimmune

- A. Rheumatic diseases – Including lupus, rheumatoid arthritis, vasculitis, scleroderma, mixed connective disease
- B. Other – Granulomatosis with polyangiitis (Wegener's), polyarteritis nodosa, sarcoidosis, IBD (Crohn's, ulcerative colitis), Whipple's Disease, giant cell arteritis, Behcet's disease, rheumatic fever

Etiology.....

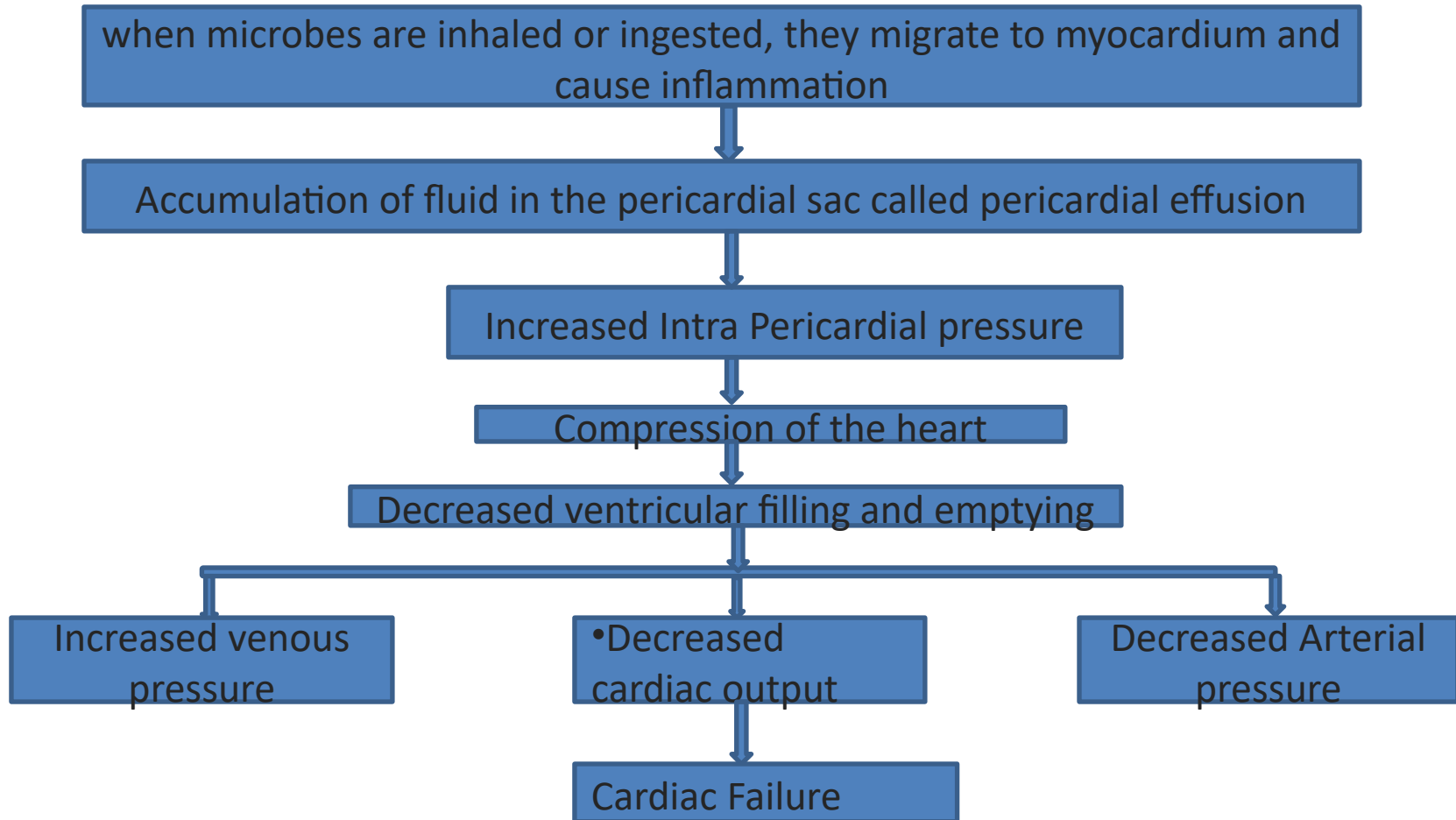
6. Drugs

- Pericarditis can also develop from a drug- induced lupus syndrome caused by medications including procainamide, hydralazine, methyldopa, isoniazid, mesalazine, and reserpine.
- Doxorubicin: The anthracycline antineoplastic agents, such as doxorubicin and cyclophosphamide, have direct cardiac toxicity and can cause acute pericarditis
- Penicillin : Penicillin and cromolyn sodium, induce pericarditis through a hypersensitivity reaction
- Methysergide: Methysergide antimigraine drug belongs to the group of medicines known as ergot alkaloids. It causes constrictive pericarditis through mediastinal fibrosis

7. Metabolic

- A. Hypothyroidism - Primarily pericardial effusion
- B. Uremia
- C. Ovarian hyperstimulation syndrome

Pathophysiology



Clinical features

- Chest pain beneath the clavicle, in the neck region worsens with deep inspiration, relieved with sitting or leaning forward. It is the cardinal sign of pericarditis
- Mild fever, chills and night sweats.
- Malaise, myalgia
- Dyspnea due to constriction or cardiac tamponade
- Palpitation
- Ewart sign: Ewart's sign is a set of findings on physical examination in people with large collections of fluid around their heart (pericardial effusions). Dullness to percussion ("woody" in quality), egophony, and bronchial breath sounds may be appreciated at the inferior angle of the left scapula when the effusion is large enough to compress the left lower lobe of the lung
- Beck's triad: falling BP; rising JVP;

Clinical features

In Constrictive Pericarditis :

- Pedal edema
- Hepatomegaly
- Ascites
- JVD
- Kussmaul's sign
- Pericardial knock (early diastolic sound) heard at the apex
- Usually - no friction rub

Medical management

- ASA or tylenol Acetaminophen decreases fever and pain , but does not help inflammation. Adult dosing is 2 regular strength (325 mg) every 4 hours or 2 extra-strength (500 mg) every 6 hours. Maximum dose is 4,000 mg per day.
- Aspirin or NSAIDs are recommended as first-line therapy for acute pericarditis with gastroprotection. Commonly used NSAID regimens include : Ibuprofen — Depending on the severity of the pericarditis and individual medication response, a dose of 400 to 800 mg of ibuprofen three times daily is usually adequate for symptom relief. Ibuprofen can be the preferred NSAID because of its rare side effects, favorable impact on coronary artery blood flow, and large

Medical management

- Aspirin — Aspirin can be given at a dose of 750 to 1000 mg every six to eight hours followed by gradual tapering every week for a treatment period of three to four weeks.
- Corticosteroids Corticosteroids are strong medications that fight inflammation. Your doctor may prescribe a corticosteroid such as prednisone if your symptoms don't get better with other medications, or if symptoms keep returning.
- Colchicine anti-inflammatory agent It is recommended as first-line therapy for acute pericarditis as an adjunct to aspirin/NSAID therapy. You should not take this drug if you have liver or kidney disease
- Indomethacin — Indomethacin (NSAID) can be administered at a dose of 50 mg three times daily for one to two weeks followed by slow tapering But commonly it is not recommended due to its adverse effects
-

Medical management

- Penicillin - for Bacterial infection
- ACE Inhibitors - relax the blood vessels in the heart and help blood flow more easily
- Beta-blockers are avoided because it decreases the strength of ventricular contraction (have a negative inotropic effect)
- Anticongestive measures such as diuretics And Inotropics agents (Inotropic agents such as milrinone, digoxin, dopamine, and dobutamine are used to increase the force of cardiac contractions.)
- Anti-anxiety medication (Alprazolam
Diazepam ,Estazolam ,Flurazepam)
- Proton pump inhibitors (Omeprazole, Pantoprazole)

Surgical management

- **Pericardiocentesis:-** is the aspiration of fluid from the pericardial space that surrounds the heart.
- **Pericardial window** a small opening made in the pericardium, may be performed to allow continuous drainage into the chest cavity.
- **Percutaneous balloon pericardiotomy :-** is a procedure done to drain excess fluid in the sac around the heart. The procedure uses a long thin tube with a balloon attached. During PBP, a doctor inserts a needle through the chest wall and into the tissue around the heart. Once the needle is inside the pericardium, the doctor removes it and replaces it with a long, thin tube called a catheter. This tube has an inflatable balloon at its tip. Repeated inflation of the balloon creates a small hole or “window” in the pericardium. When the hole is large enough, the doctor removes the catheter and balloon replaces them with a new catheter for final draining. This allows fluid to drain out of the pericardium, which improves heart function

Surgical management

- Pericardiectomy may be necessary to release both ventricles from the constrictive and restrictive inflammation and scarring. Pericardiectomy is performed through a median sternotomy, an incision through the breastbone (sternum) in the middle front part of the ribs that allows the surgeon to reach the heart. The surgeon will remove the pericardium from the heart, wire the breastbone and ribs back together and close the incision with stitches.